Enhancing Clinician Trust through a Trust-Enabling Framework for Clinical RAG-Based LLM Assistants interfacing with Internal Healthcare Protocols

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Abstract

Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) systems are increasingly explored in healthcare to support access to clinical protocols, guidelines, and internal documentation. However, improved information retrieval alone does not resolve concerns around trust, transparency, safety, accountability, and human oversight in clinical use, which were identified in the literature review and recognized during expert consultation in hospital context. This study investigates how clinician trust and transparency can be supported through the design and operationalization of a trust-enabling framework for clinical RAG-based LLM assistants. Following a Design Science Research approach, the study combines a literature review, expert consultation, ethical evaluation, requirement formulation, architecture design, prototyping, and formative validation. The resulting Clinician Trust Framework functions as a trust-enabling layer around a RAG-based LLM system interfacing with internal clinical protocols. It combines source provenance, evaluation scores, warning generation, trace logging, dashboard monitoring, behavioural safeguards, and feedback capture to make AI-generated answers more inspectable and subject to review. A proof-of-concept implementation demonstrates how trust-supporting mechanisms can be operationalized around a clinical RAG-based LLM assistant. Formative validation consisted of a scenario-based one-to-one expert validation with a Head Research Nurse and group validation with nurses. The expert validation indicated that the trust-supporting mechanisms can support informed judgement and trust calibration, while the group validation showed that nurses recognized the value of source visibility, reliability information, and answer explanation, but also identified limitations in source presentation, terminology, and interface design.

1 Introduction

Healthcare professionals rely on timely access to accurate information to support safe and effective patient care. As healthcare organizations continue to digitize clinical knowledge resources, there is growing interest in tools that help clinicians retrieve relevant information more efficiently from large collections of internal documentation. In many clinical settings, nurses access medical protocols through centralized documentation systems that contain organizational and clinical guidelines. Although these systems provide access to formal protocol information, finding the right protocol can be time-consuming, especially when users need to navigate multiple related documents.

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Retrieval-Augmented Generation (RAG)-based Large Language Model (LLM) assistants offer a promising way to improve access to internal protocol knowledge. Such systems allow users to ask questions in natural language, retrieve relevant documents, and generate answers grounded in those retrieved sources [32, 12, 13]. In a protocol-driven clinical context, this could reduce the effort required to locate relevant documentation and make internal medical knowledge more accessible during clinical work.

However, introducing RAG-based LLM assistants in clinical environments also creates a technically and organizationally complex problem. The system does not only depend on the language model, but also on the quality of the document corpus, chunking strategy, embeddings, vector retrieval, prompt construction, answer generation, evaluation logic, safety mechanisms, logging, monitoring, user feedback, and governance processes. If these elements are not made visible and reviewable, clinicians may receive an answer without sufficient insight into where it came from, whether the retrieved sources were appropriate, how well the answer was grounded, or whether the system behaved safely within its intended scope.

This study addresses that problem by designing and operationalizing the *Clinician Trust Framework*, a proof-of-concept trust-enabling layer around a clinical RAG-based LLM assistant interfacing with internal healthcare documentation. The framework does not aim to introduce a new retrieval or generation method. Instead, it focuses on making RAG-based interactions more transparent, inspectable, traceable, monitorable, and open to human review. It combines source provenance, evaluation scores, warning generation, trace logging, dashboard monitoring, behavioural safeguards, and feedback capture to support calibrated trust in AI-generated protocol answers. The central objective of this study is to investigate how clinician trust and transparency can be supported through concrete technical and governance mechanisms around a clinical RAG-based LLM assistant.

The remainder of this paper is structured as follows. Section 2 presents the background and related work on clinical RAG, trust, transparency, monitoring, and governance. Section 3 describes the research methodology. Section 4 presents the framework architecture, proof-of-concept implementation, and formative validation findings. Section 5 discusses the results, limitations, and implications, and Section 6 concludes the study.

2 Related Work

Retrieval-Augmented Generation is an approach in which an LLM retrieves relevant information from external knowledge sources and uses this information as context when generating a response [32]. In clinical settings, RAG can make LLM-based systems more useful by reducing reliance on the model's internal parameters alone. Instead of generating answers only from general model knowledge, a RAG-based system retrieves relevant clinical documents and uses them to produce a more context-grounded response.

This grounding is particularly relevant in healthcare, where generated answers should correspond to validated documentation rather than general or unsupported model output. In protocol-driven clinical work, answers should not only sound plausible, but must correspond to the correct internal protocol for the specific care situation. Clinicians must therefore be able to assess which protocol was retrieved, whether that protocol is appropriate, and whether the generated response is adequately grounded in the retrieved source [14, 3].

Although RAG can improve the grounding of generated answers, retrieval alone does not make the final response reliable, transparent, or safe. A basic RAG workflow retrieves documents and uses them to generate an answer, but it does not necessarily expose whether the retrieved evidence was sufficient, how the answer was evaluated, whether warnings were triggered, or how the interaction can be reviewed afterwards. Figure 1 illustrates the difference between a basic RAG assistant and the proposed Clinician Trust Framework.

The deployment of LLM-based systems in clinical environments introduces several socio-technical challenges. Healthcare professionals have emphasized that AI systems must be transparent, inter-

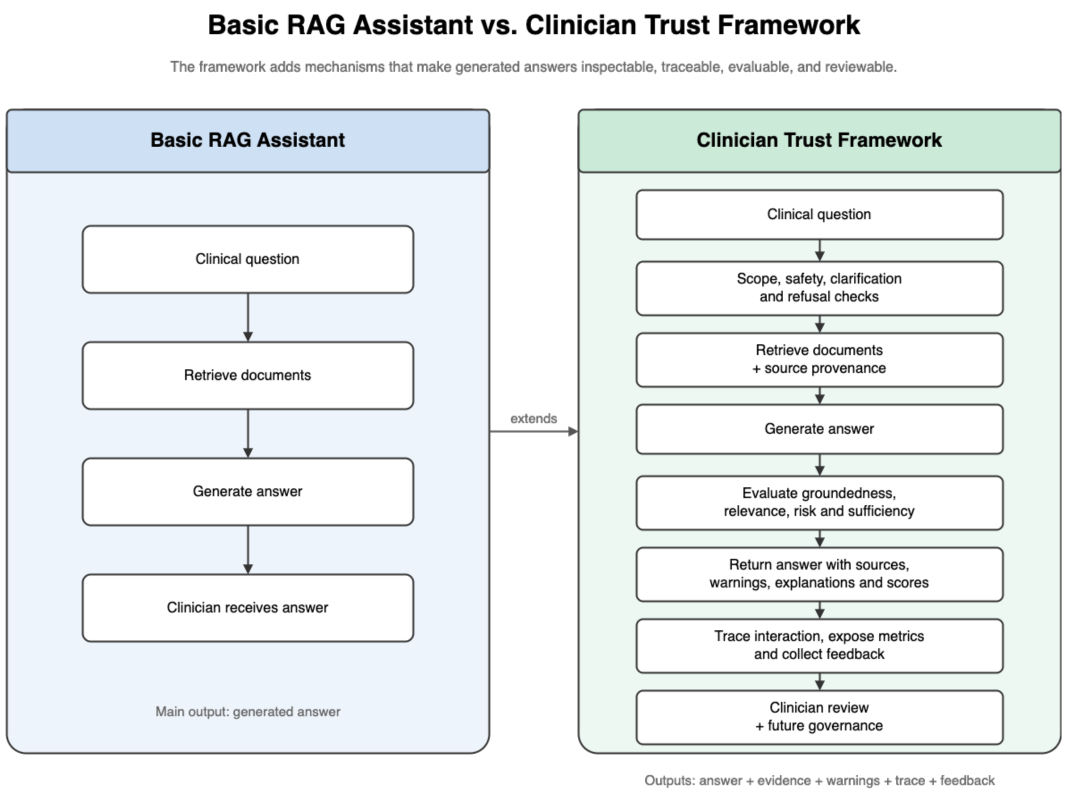

Figure 1: Comparison between a basic RAG assistant and the proposed Clinician Trust Framework. While basic RAG focuses on retrieving documents and generating an answer, the Clinician Trust Framework adds safety checks, source provenance, evaluation scores, warnings, trace logging, monitoring, feedback, and review mechanisms to support clinician trust calibration.

pretable, and accountable before they can be responsibly integrated into clinical workflows [14, 30]. Current research on clinical LLM implementation identifies recurring risks, including incorrect or incomplete recommendations, limited explainability, privacy concerns, bias, unsafe outputs, and unclear responsibility when errors occur [1, 3, 20]. These risks are especially relevant in healthcare, where system outputs may influence professional judgement, care coordination, or downstream clinical actions.

Trust is therefore a central concern. In clinical AI, trust should not be treated only as a subjective attitude or as a simple measure of user satisfaction. Instead, it should be understood as a calibrated socio-technical outcome. Clinicians should be able to use an AI-generated answer when it is well supported and appropriate, but should also be able to question, verify, override, or reject the answer when it is uncertain, weakly grounded, unsafe, or outside the system's intended scope [27, 4]. Such calibrated trust requires visible evidence around the answer, including retrieved sources, provenance information, explanation cues, warning indicators, and feedback mechanisms.

Existing work on trustworthy AI, clinical AI evaluation, MLOps, and LLMOps provides useful principles for addressing these challenges. Clinical AI guidance emphasizes early-stage evaluation, human oversight, safety, and workflow relevance [31]. MLOps and observability research highlights the importance of monitoring deployed models, tracking data and model versions, detecting drift, and maintaining reproducible records of system behaviour [22, 25]. Governance frameworks and regulatory discussions further stress the need for documentation, auditability, risk management, post-deployment monitoring, and clearly assigned accountability [15, 17, 6].

For RAG-based LLM systems, these concerns extend beyond the model itself. They also include the retrieval corpus, vector index, prompts, retrieved passages, generated answers, user feedback, and subsequent human actions [29, 23]. Recent work on RAG trustworthiness highlights the importance of responsible source use, grounded attribution, refusal behaviour, and the presentation of evidence, citations, and trust cues [29, 23, 9]. These studies support the importance of source-grounded and explainable RAG, but they do not fully address how such mechanisms can be integrated with tracing, monitoring, feedback, and governance in a clinical proof-of-concept architecture.

A gap remains between high-level trustworthy AI principles and their operationalization in practical clinical RAG architectures. Existing guidance often describes what should be achieved, including transparency, accountability, explainability, fairness, human oversight, and post-deployment monitoring [3, 2, 6]. However, these principles are less often translated into concrete monitoring dimensions, measurable indicators, logging mechanisms, warning systems, feedback loops, and governance workflows for RAG-based LLM assistants.

Moreover, monitoring is often treated as a backend engineering activity focused on uptime, latency, or model performance, while its connection to clinician trust, clinical workflow, human oversight, and institutional accountability remains underdeveloped [22, 18, 17, 25]. This study addresses that gap by designing and implementing the Clinician Trust Framework as a trust-oriented layer around a clinical RAG pipeline. The framework operationalizes trust-supporting mechanisms through source provenance, answer evaluation, warning generation, trace logging, dashboard monitoring, behavioural safeguards, and clinician feedback.

3 Methodology

The study follows a Design Science Research approach to design, implement, and formatively validate a trust-enabling framework for clinical RAG-based LLM assistants. Design Science Research is appropriate because the objective is not only to understand a problem, but also to construct and validate an artifact that addresses that problem in a practical context [19]. In this study, the artifact is the Clinician Trust Framework: a trust-enabling layer around a clinical RAG-based LLM assistant that combines monitoring, evaluation, traceability, feedback, and governance mechanisms.

This methodological design is consistent with prior applied AI and RAG evaluation work in which literature synthesis, expert input, prototype development, and formative evaluation are combined to construct and assess a practical artifact. For example, design-oriented RAG evaluation work has used literature synthesis, expert feedback, field trials, logging, and modular evaluation mechanisms to develop more transparent RAG assessment frameworks [26, 21]. Similarly, MLOps research has combined literature review, tool review, and expert input to derive operational principles, architecture components, and workflows for production-oriented machine learning systems [8]. In the clinical AI context, prior work also emphasizes the importance of clinician perspectives, human oversight, expert review, and multidimensional evaluation when assessing the trustworthiness and usefulness of AI systems [14, 30, 7, 11].

The methodology was structured around four main activities. First, evidence was collected from literature, expert consultation, and ethical evaluation to identify trust-related requirements. Second, these requirements were translated into a framework architecture. Third, the architecture was implemented as a proof-of-concept prototype. Fourth, the prototype was formatively validated with clinical users to assess whether its trust-supporting mechanisms were understandable, useful, and relevant in practice.

3.1 Research Design

The research was conducted as an applied design-oriented study in a hospital context. The goal was to develop and evaluate a framework that operationalizes clinician trust and transparency around a RAG-based LLM assistant interfacing with internal healthcare protocols. The Development Oriented Triangulation (DOT) framework was used as a practical guide for combining different types of research activities, including literature review, expert consultation, design, prototyping, and validation [5]. In this study, literature review corresponds to the library strategy, expert consultation to the field strategy, architecture design to the workshop strategy, prototyping to the lab strategy, and formative validation to the showroom strategy.

3.2 Literature Review

The literature review was used as a design input rather than as a standalone systematic review. Its purpose was to identify recurring concerns, requirements, monitoring dimensions, and governance principles relevant to trustworthy clinical RAG-based LLM systems. This approach aligns with design-oriented AI engineering studies, where literature synthesis is used to derive requirements and architectural choices before prototyping and validation [8, 26].

The literature review focused on clinical AI trust, RAG-based LLM systems, healthcare AI governance, LLMOps, MLOps, observability, auditability, transparency, and human oversight. Sources were selected when they addressed clinical use of LLMs, trust in AI-based clinical decision support, RAG-specific risks, monitoring and observability, governance and accountability, auditability, or healthcare AI evaluation.

The review primarily focused on literature from 2020 onward, because RAG-based LLM systems, LLMOps, and clinical LLM implementation are rapidly developing areas in which recent work is most relevant. Earlier sources were included when they provided foundational or regulatory context for broader issues such as trustworthy AI, clinical AI governance, transparency, auditability, and human oversight. In practice, this meant that foundational and regulatory sources from 2016 onward were considered where relevant. Sources were identified through academic databases and repositories, including PubMed, ScienceDirect, JMIR, arXiv, medRxiv, MDPI, Nature, and DOI.org. Search terms combined healthcare, technical, and governance-oriented concepts, including ‘large language models in healthcare”, ‘RAG”, ‘LLMOps”, ‘MLOps”, ‘AI governance”, ‘AI transparency”, ‘AI explainability”, ‘system monitoring”, ‘clinical AI trust”, ‘trustworthy AI”, and “responsible AI”.

Initial screening was based on titles and abstracts. Sources were included when they addressed at least one of the following areas: clinical use of LLMs, trust in AI-based clinical decision support, RAG-specific risks, monitoring and observability, governance and accountability, auditability, or healthcare AI evaluation. The selected literature was then analysed in relation to the research questions to identify recurring requirements, monitoring dimensions, trust indicators, and governance concerns. This process provided the initial conceptual basis for the framework.

3.3 Expert Consultation

Expert consultation was conducted to assess whether the trust-related concerns identified in the literature were recognizable and actionable within the hospital context. Stakeholders were selected to represent complementary perspectives relevant to clinical RAG-based LLM deployment: clinical use, ethics, legal responsibility, implementation feasibility, and user-interface design. The consultations included the Head Research Nurse of the geriatrics department, an ethicist, a legal expert, an IT Administrator and an Industrial Design master's graduation student.

The consultations were structured around three discussion themes: the relevance of literature-derived risks within the Hospital, the identification of additional context-specific risks or requirements, and the translation of these concerns into framework requirements. In addition, multiple design-focused meetings were held with the Industrial Design master's student to translate trust-supporting mechanisms into UI/UX components, such as source visibility, warning presentation, reliability information, feedback interaction, visual hierarchy, and clarity of answer explanation. Notes from the consultations were compared with the literature findings and used to refine the requirement areas, including transparency, traceability, monitoring, human oversight, governance, privacy, responsibility, clinician autonomy, and interface understandability.

3.4 Ethical Evaluation

An ethical evaluation was conducted to examine the broader implications of deploying and monitoring a clinical RAG-based LLM assistant (Appendix E). This was necessary because the framework does not only evaluate technical performance. It also introduces logging, traceability, feedback capture, and governance mechanisms that may affect clinicians, patients, and organizational responsibilities.

The Technology Impact Cycle Tool (TICT) was used as the main ethical reflection instrument [28]. TICT supports structured reflection on the social, ethical, and practical consequences of technology. The ethical evaluation focused on legal responsibility, privacy, clinician autonomy, data quality, fairness and bias, transparency, and communication. The results of this ethical evaluation were used as design input for the framework requirements and limitations.

3.5 Requirement Derivation

The framework requirements were developed by systematically translating evidence from the literature review, expert consultations, and ethical evaluation into design requirements. First, trust-related concerns were extracted from the literature review, expert consultations, and ethical evaluation. These concerns were then grouped into recurring requirement areas, including transparency, provenance and traceability, monitoring, human oversight, governance, safety, privacy, and interface understandability. Each area was translated into one or more design requirements by asking what the system would need to expose, record, evaluate, warn about, or support in order to make RAG-based answers more inspectable and reviewable.

This process resulted in requirements for end-to-end provenance, transparent answer presentation, continuous monitoring, versioning of relevant artifacts, risk monitoring, feedback capture, regulatory traceability, and stakeholder involvement. The same synthesis process was also used to define monitoring dimensions, including retrieval quality, grounding of generated claims, context sufficiency, uncertainty handling, refusal behaviour, toxicity, auditability completeness, and human feedback. The resulting requirements formed the design input for the Clinician Trust Framework architecture and proof-of-concept implementation.

3.6 Artifact Design

The Clinician Trust Framework was designed by translating the derived requirements into an architecture for a clinical RAG-based LLM assistant. Architecture sketching was used to reason about the system from several viewpoints, including context, container, component, runtime, API, data, monitoring, and deployment views. These viewpoints clarified how the framework relates to users and supporting services, how components are deployed, how the RAG workflow operates, how monitoring data is captured, and how feedback and traceability are handled.

The resulting architecture positioned the Clinician Trust Framework as a trust-enabling layer around a clinical RAG-based LLM assistant. It included components for document ingestion, vector retrieval, LLM-based answer generation, safety and clarification checks, evaluation scoring, warning generation, tracing, metrics, dashboards, and user feedback capture. The architecture was intentionally scoped as a proof-of-concept rather than as a production-ready hospital AI system.

3.7 Proof-of-Concept Implementation

A proof-of-concept implementation was developed to examine whether the architectural design could be operationalized in a working technical artifact. The prototype implemented protocol-grounded question answering, source retrieval, behavioural safeguards, scoring, warning generation, trace logging, metrics exposure, and feedback capture.

The purpose of the prototype was not to optimize the underlying LLM or prove clinical effectiveness. Instead, the prototype was used to demonstrate that trust-supporting mechanisms such as traceability, evaluation, warning generation, feedback, and oversight can be integrated around a clinical RAG-based LLM assistant.

3.8 Formative Validation Design

The validation design was informed by early-stage clinical AI evaluation and LLM evaluation literature, which emphasizes that systems intended for clinical contexts should be assessed not only through technical performance, but also through expert judgement, usability, clinical relevance, transparency, and human oversight [31, 7, 30]. Because automatic evaluation scores alone are insufficient to assess the practical value and trustworthiness of generated answers, the validation combined survey responses, scenario-based interaction, observations, and qualitative feedback [24, 11, 30].

The validation consists of two complementary formative evaluation activities: 1) a one-to-one expert scenarion-based validation and 2) a scenario-based group validation with nurses. Both activities focus on assessing whether the framework's trust-supporting mechanisms are understandable, useful, and clinically relevant when evaluating AI-generated protocol answers.

Both validation activities follow the same procedure (Appendix C). Participants first complete a baseline questionnaire capturing professional background, prior AI experience, and attitudes toward AI in clinical practice. They then receive a short introduction to the prototype and work through the same set of protocol-based scenarios. The scenarios require participants to retrieve protocols, inspect sources, interpret AI-generated answers, use feedback mechanisms, and respond to situations involving uncertainty or terminology mismatches. After completing the scenarios, participants fill in the same post-use questionnaire covering usability, perceived trustworthiness, source visibility, and the usefulness of the trust-supporting information.

The one-to-one expert validation was conducted with the Head Research Nurse of the geriatrics department. The expert completed the same baseline questionnaire, scenarios, and post-use questionnaire as the group participants. The individual format additionally allowed for in-depth discussion of the AI-generated answers, the trust-supporting information, and how these elements influenced interpretation, confidence, and decision-making.

The group validation was conducted with nurses using the same questionnaires and scenarios as the expert validation. This ensured that feedback from the individual expert session and the group sessions could be compared directly while still allowing the collection of broader perspectives on usability, source inspection, answer interpretation, and the practical value of the framework's trust-supporting mechanisms.

The evaluation is formative rather than summative. The objective is not to demonstrate clinical effectiveness or statistical generalizability, but to determine whether the framework mechanisms are understandable, practically relevant, and useful for supporting answer inspection and trust calibration. Combining an expert validation with a group validation provides both detailed qualitative feedback on clinical interpretation and broader feedback on usability, source visibility, answer presentation, and the practical use of trust-supporting mechanisms.

The group validation dataset contained 16 baseline responses and 10 post-use responses. After cleaning participant identifiers, 10 responses could be paired across the baseline and post-use questionnaires. Quantitative responses were analyzed descriptively because the validation was formative and not statistically powered. Qualitative comments and observations were used to interpret the survey patterns and identify refinements.

4 Results

4.1 Trust-Supporting Requirements

The first outcome of this study is a set of requirements for a trust-enabling framework for clinical RAG-based LLM assistants. The requirements analysis showed that clinician trust cannot be addressed through answer accuracy alone. Instead, the system must make each interaction inspectable, reconstructible, and open to review. This aligns with prior work showing that trust in clinical AI depends on transparency, explainability, privacy protection, ethical governance, and the ability of healthcare professionals to understand and challenge system outputs [30, 9]. In RAG-based LLM assistants, this is especially important because the generated answer is shaped not only by the model, but also by retrieved documents, prompts, embeddings, retrieval settings, and runtime configuration [29, 23]. Table 1 summarizes the resulting requirement areas and their focus within the framework.

Table 1: Summary of trust-supporting framework requirements.

Requirement area	Requirement focus
Transparency	Present answers with sources, explanations, warnings, and reliability-related information.
Provenance and traceability	Link each answer to the query, retrieved sources, prompt, model configuration, evaluation scores, feedback, and trace metadata.
Evaluation and monitoring	Capture reliability, warning frequency, and user feedback.
Behavioural safeguards	Support off-topic detection, clarification logic, refusal behaviour, toxicity checks, and warning generation when information is insufficient or unsafe.
Human review and feedback	Support source inspection, feedback capture, verification, escalation, and clinician override.
Governance and auditability	Enable audit trails, lifecycle monitoring, incident review, source maintenance, threshold refinement, and future system improvement.

The resulting requirements reflect recommendations from clinical MLOps and AI governance literature, which emphasize lifecycle monitoring, versioning, audit trails, human-in-the-loop governance, incident response, and clearly defined operational responsibilities for deployed healthcare AI systems

[25, 15, 6]. They also align with work on LLM audit trails and multidimensional evaluation, where accountability depends on capturing the conditions under which an output was produced and avoiding reliance on a single performance score [16, 11, 24].

4.2 Clinician Trust Framework Architecture

The second outcome is the Clinician Trust Framework architecture. The framework was designed as a proof-of-concept architecture around a clinical RAG-based LLM assistant. Its purpose is not only to answer questions over internal clinical protocol documents, but also to expose the evidence and monitoring information needed to inspect how an answer was produced. This positioning is consistent with responsible AI guidance in healthcare, which emphasizes transparency, accountability, human oversight, and responsible integration into clinical workflows [3, 2]. It also reflects RAG-specific concerns that source provenance, retrieved evidence, and citation quality are central to trustworthiness in systems that combine retrieval and generation [29, 23].

Figure 2 presents the framework as a layered structure around a clinical RAG-based LLM assistant. The clinician-facing layer contains the answer, sources, warnings, explanations, and feedback options. The trust evidence layer captures provenance and evaluation signals. The observability layer records traces, metrics, and dashboard information, while the governance layer supports audit, escalation, accountability, and future improvement. Together, these layers connect immediate answer inspection with longer-term monitoring and governance.

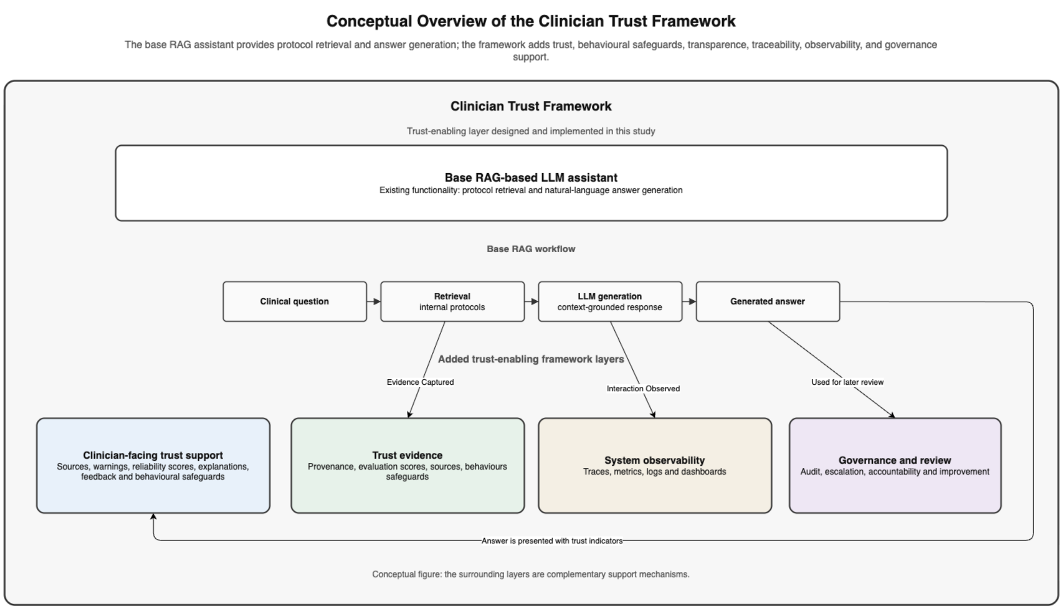

Figure 2: Conceptual overview of the Clinician Trust Framework. The base RAG-based LLM assistant provides protocol retrieval and answer generation, while the framework designed in this study adds clinician-facing trust support, trust evidence, system observability, and governance-oriented review. The surrounding layers are complementary support mechanisms and should not be interpreted as a sequential workflow.

Figure 3 presents the high-level architecture of the framework. The diagram positions the framework as a trust-oriented layer around the RAG assistant rather than as a standalone question-answering system. It shows how user questions move through the backend, where safety checks, retrieval, prompt construction, generation, evaluation, warning generation, tracing, metrics, and feedback handling are coordinated. It also shows how the backend connects to the vector store, model services, tracing platform, metrics endpoint, and monitoring dashboard.

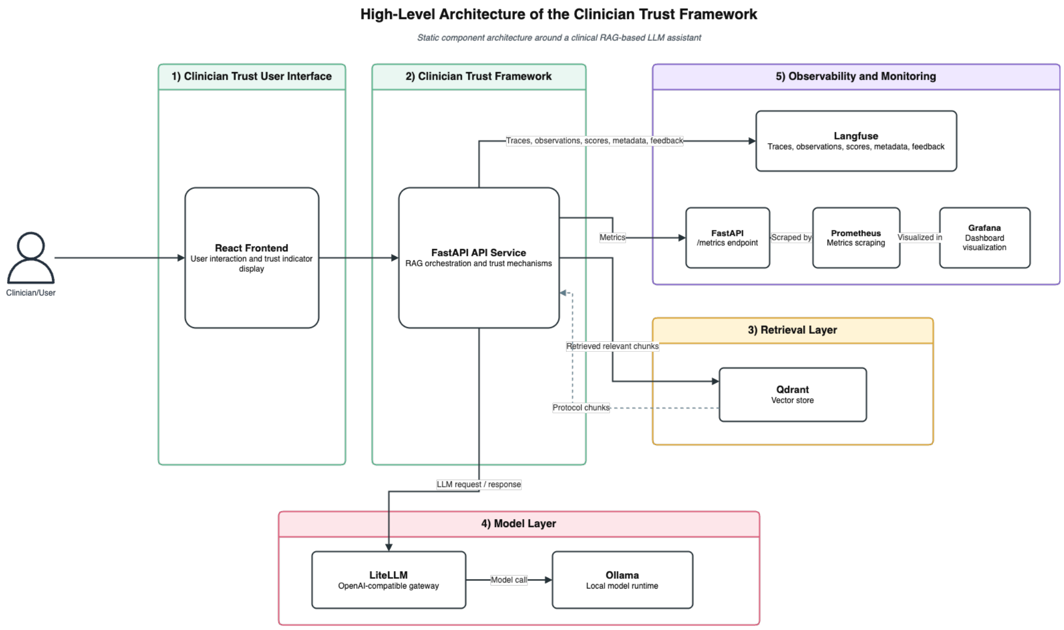

Figure 3: High-level architecture of the Clinician Trust Framework around a clinical RAG-based LLM assistant.

The architecture should therefore be understood as a trust-enabling layer around a RAG-based LLM assistant. It extends the basic RAG pipeline with mechanisms for source visibility, evaluation scoring, warning generation, traceability, monitoring, feedback capture, and human review. These mechanisms are intended to support calibrated trust: clinicians should be helped to rely on answers when they are well grounded, but also to question or reject answers when evidence is weak, missing, unsafe, or outside the system's intended scope [9, 23].

4.3 Proof-of-Concept Implementation and Functionality

The architecture was implemented as a local proof-of-concept to demonstrate how the framework's trust-supporting mechanisms could be operationalized in a working technical artifact. The implementation was not intended to optimize model performance or function as a production-ready clinical system. Instead, it was used to test whether a protocol-grounded RAG assistant could be extended with reviewable evidence, monitoring signals, and feedback mechanisms.

The proof-of-concept consists of several main components. A FastAPI backend coordinates the request flow, including behavioural safeguards, retrieval, generation, evaluation, warning generation, tracing, metrics, and feedback. Qdrant stores embedded protocol chunks and enables semantic retrieval. LiteLLM provides an OpenAI-compatible gateway to model backends, while Ollama supports local model execution. Langfuse captures traces, observations, generations, metadata, evaluation scores, and feedback. Prometheus-compatible metrics are exposed through the application, while Grafana is used to visualize operational and trust-related monitoring signals. These choices align with LLMOps literature that frames deployment, observability, monitoring, evaluation, and lifecycle control as central concerns for LLM-based systems in production-like environments [18] and are operationalized using Langfuse documentation [10].

The implemented workflow starts with the ingestion of clinical protocol documents, which are chunked, embedded, and indexed in the vector store. When a user submits a question, the backend first performs safety, scope, and clarification checks. It then retrieves relevant protocol chunks, constructs the prompt, generates an answer, evaluates the response, stores trace and metric data, and returns the answer together with supporting information such as sources, scores, warnings, and trace metadata.

Figure 4 presents the runtime interaction flow of the proof-of-concept Clinician Trust Framework. It shows how a clinician question is submitted to the system through the `/ask` endpoint, processed through safety, scope, clarification, and refusal checks, enriched with retrieved protocol context, and sent to the model runtime for answer generation. The generated answer is then evaluated, traced, and enriched with supporting information such as sources, reliability signals, warnings, and trace metadata before being returned to the clinician as a reviewable answer package.

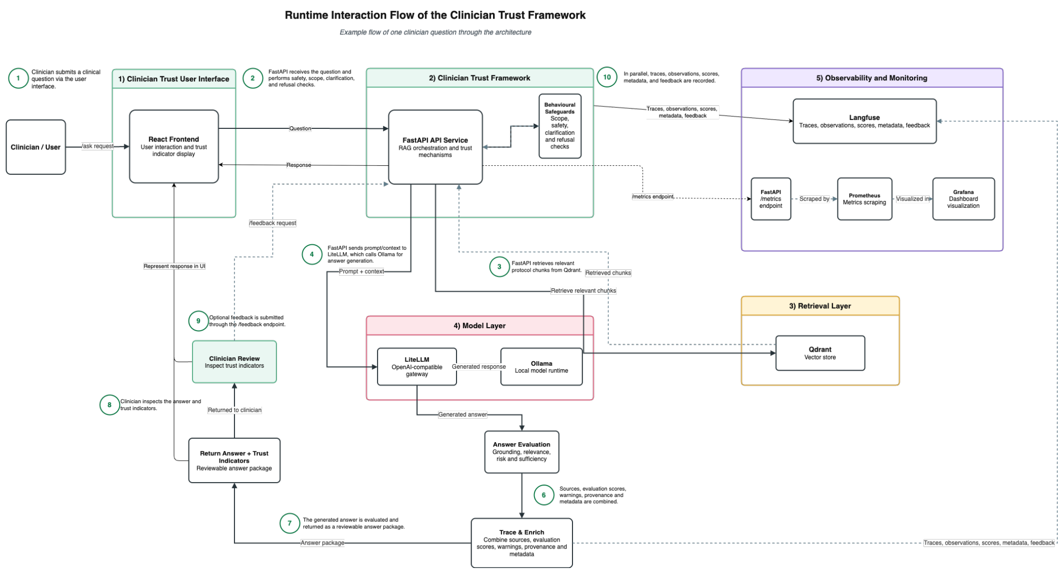

Figure 4: Runtime flow of the proof-of-concept Clinician Trust Framework, from document ingestion and question answering to evaluation, tracing, monitoring, and feedback.

Figure 4 also shows how the framework supports feedback and monitoring after the answer has been returned. If the answer is unclear, incorrect, or incomplete, the clinician can provide feedback through the `/feedback` endpoint. In parallel, trace data, evaluation scores, warnings, and operational metrics are exposed through the `/metrics` endpoint and made available through monitoring dashboards for operational oversight and performance review. Together, these endpoints support an operational cycle from question answering to answer inspection, feedback capture, monitoring, and later review.

The proof-of-concept implements behavioural safeguards, including off-topic detection, clarification logic, refusal behaviour, toxicity checks, and warning generation when information is insufficient, unsafe, or outside the system's intended scope. Their purpose is not to guarantee clinical correctness, but to make uncertainty and insufficient evidence visible during use. This aligns with RAG trustworthiness research on grounded attribution and appropriate refusal when available evidence is insufficient [23]. It also reflects clinical AI guidance that systems should support human oversight rather than encourage automation reliance [31, 3].

4.4 Example Use Case: Inspecting a Protocol Answer

To illustrate how the framework supports trust calibration in practice, Figure 5 shows an example interaction from a protocol-retrieval scenario. In this scenario, a nurse asks which protocol should be used when a patient has residual urine after micturition and a catheter needs to be inserted. This type of question is suitable for demonstrating the framework because several related protocols may be retrieved, while only one protocol may be most appropriate for the specific situation.

The prototype does not only return a generated answer. It also exposes supporting information that helps the clinician inspect the response, including source information, reliability-related signals, warnings, and feedback options. In this way, the interaction is designed to support verification and review rather than passive reliance on the generated output.

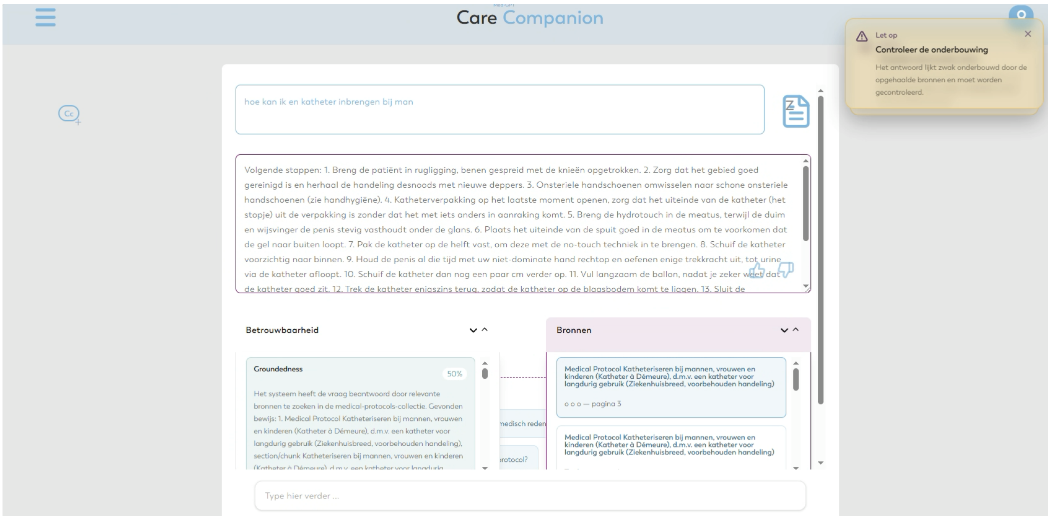

Figure 5: Example use case showing how the prototype supports inspection of a protocol answer. The generated answer is presented together with source information, reliability or monitoring information, and feedback options, allowing the clinician to verify rather than passively accept the response.

Figure 5 shows how the framework presents the generated answer together with evidence and review cues. The answer can be checked against the retrieved sources, while the reliability information, warnings, and feedback option provide additional support for judging whether the response should be accepted, questioned, or corrected. Table 2 summarizes the main trust-supporting elements visible in this example and their role in the interaction.

Table 2: Trust-supporting elements shown in the example use case.

Framework requirement area	**Example use-case implementation**
Transparency	The answer is presented with source information, an explanation of the response, reliability-related information, and warning cues to help the clinician interpret the output.
Provenance and traceability	The clinician can inspect the protocol source used for the answer, and interaction data can be linked to the question, answer, sources, warnings, and feedback.
Evaluation and monitoring	Reliability-related information is displayed to provide insight into the quality and grounding of the generated answer.
Behavioural safeguards	Warning signals highlight uncertainty, missing evidence, or potential risks associated with the generated response, as well as follow-up suggestions for further clarification.
Human review and feedback	The clinician can review the supporting source material and provide feedback on incorrect, unclear, or incomplete answers.
Governance and auditability	Interaction information can be retained for later review, including the question, answer, sources, evaluation signals, warnings, and feedback.

The example shows that the generated answer is not presented as a standalone recommendation, but as part of a reviewable interaction. The clinician can inspect sources, consider reliability information and warnings, and provide feedback when the answer is unclear or incorrect. Additional dashboard and traceability views are included in Appendix D.

4.5 Formative Validation of Trust-Supporting Mechanisms

Table 3 shows that the validation findings provide support for the framework's design direction, while also identifying important refinements. The one-to-one expert validation gave the clearest indication that the trust-supporting mechanisms can improve answer assessment, especially through citations, provenance information, metrics, warnings, and explanations (Appendix A). The group validation was more mixed, but it still showed that nurses recognized the relevance of specific mechanisms such as source visibility, reliability information, and answer explanation (Appendix B).

Table 3: Mapping of framework requirement areas to implemented features and formative validation findings.

Requirement area	**Implemented features**	**Formative validation findings**	**Main refinement**
Transparency	Generated answers are presented with source information, answer explanation, reliability information, and warning cues.	The one-to-one expert validation showed improved understanding of why the answer was generated after the trust-supporting information was shown. In the group validation, participants repeatedly identified answer explanation, reliability information, and source visibility as helpful.	Improve answer structure and terminology.
Provenance and traceability	Answers are linked to retrieved sources, protocol context, evaluation scores, feedback, and trace-related metadata.	The one-to-one expert validation showed improvement in tracing the origin of information and reconstructing how the system arrived at the answer. In the group validation, source visibility was the clearest positive quantitative signal, although participants indicated that source contribution was not always clear.	Clarify source contribution.
Evaluation and monitoring	The framework exposes evaluation scores, reliability information, warning signals, trace logging, metrics, and dashboard-oriented monitoring information.	The one-to-one expert validation showed that trustworthiness scores and warnings were among the most useful features. Group participants also mentioned reliability information as useful, but the survey results show that the current presentation did not consistently increase perceived reliability or trust.	Improve metric explanation and visual presentation.
Behavioural safeguards	The prototype includes off-topic detection, clarification logic, refusal behaviour, toxicity checks, and warning generation when information is insufficient or unsafe.	The validation findings suggest that visible limitations and warnings can support more critical answer assessment. However, the group validation also showed that answer quality and unclear source use still limited confidence in the current prototype.	Clarify safeguard behaviour and warning language.

Table 3: Mapping of framework requirement areas to implemented features and formative validation findings. continued

Requirement area	Implemented features	Formative validation findings	Main refinement
Human review and feedback	The prototype includes source inspection, feedback capture, warning visibility, and support for clinician verification and override.	The one-to-one expert validation showed a shift from possible rejection or escalation toward acceptance with verification, suggesting informed reliance rather than blind acceptance. During the group validation, participants used feedback when prompted and expressed a need to inspect the original protocol.	Integrate review actions more clearly into the workflow.
Governance and auditability	Interactions are captured through trace logging, evaluation scores, feedback, warning data, and monitoring endpoints.	The one-to-one expert validation ranked audit and logging information lower for immediate answer assessment, suggesting that these features may be more relevant for retrospective review and governance than for point-of-use clinical interpretation.	Separate clinician-facing and governance-facing information.
Supports calibrated trust	The framework combines transparency, traceability, evaluation, safeguards, feedback, and governance-oriented logging to make both evidence and limitations more visible.	The one-to-one expert validation showed increased trust and decision confidence while preserving verification. The group validation did not show a general increase in trust, but qualitative responses suggest that participants became more critical of answer quality, source use, and reliability.	Improve answer quality and calibration cues.

Table 3 shows that the framework claims were only partially fulfilled in the current proof-of-concept. The mechanisms were implemented and recognized as relevant, but their presentation was not yet consistently clear or convincing for all users. In particular, the validation indicates that source contribution, reliability terminology, answer structure, and feedback integration require further refinement. The findings therefore support the relevance of the trust-enabling mechanisms, but not yet a strong claim that the prototype reliably improves clinician trust in practice.

5 Discussion

This study investigated how clinician trust and transparency can be supported through the design and operationalization of a trust-enabling framework for clinical RAG-based LLM assistants. The resulting Clinician Trust Framework demonstrates that trust-supporting concepts such as transparency, provenance, traceability, evaluation, warning generation, and feedback can be translated into concrete runtime mechanisms around a RAG pipeline. Its main contribution is therefore not the development of a clinical question-answering assistant, but the design of a trust-enabling layer that makes the behaviour of such an assistant more inspectable, monitorable, and open to human review.

The results support the view that trust in clinical LLM systems cannot be addressed through answer quality alone. Although RAG can improve factual grounding by retrieving relevant source material before answer generation, it does not automatically make the final response transparent, safe, or accountable. Retrieved sources may be incomplete, outdated, weakly related to the question, or incorrectly used by the model [14, 3]. The framework therefore extends the basic RAG workflow with mechanisms that expose retrieved evidence, evaluation scores, warnings, trace identifiers, and user feedback. This aligns with prior work arguing that trustworthy healthcare AI requires transparency, explainability, human oversight, and governance mechanisms that make system behaviour understandable and contestable in practice [30, 20, 17, 6].

The proof-of-concept also implements behavioural safeguards that support this trust-enabling role. These include off-topic detection, clarification logic, refusal behaviour, toxicity checks, and warning generation. The purpose is not to guarantee clinical correctness, but to make uncertainty, risk, and insufficient evidence more visible during use. For example, the system can avoid answering when no relevant documents are retrieved, ask for clarification when a question is ambiguous, or warn when groundedness is low. This aligns with RAG trustworthiness research that highlights the importance of grounded attribution, and refusal behaviour [23]. It also reflects clinical AI guidance that systems should support human oversight rather than encourage automation reliance [31, 17].

A key effect of the framework is that it shifts the clinical interaction from evaluating an isolated generated answer to evaluating a broader interaction record. In the implemented proof-of-concept, each answer is accompanied by supporting information, including sources, evaluation scores, warnings, and trace metadata. By making these artifacts more visible, the framework supports post-hoc inspection and provides a stronger basis for accountability than answer-only interfaces.

The validation results provide a nuanced picture of the framework's potential. The one-to-one expert validation gave a positive indication that citations, provenance information, evaluation scores, warnings, explanations, dashboard signals, and audit/log information can support informed answer assessment. After reviewing these mechanisms, the expert reported higher trust and decision confidence while still maintaining verification as part of the intended action. This suggests that the framework can support calibrated trust under reflective conditions, rather than replacing professional judgement.

The group validation produced a more mixed picture. The quantitative results did not show a general increase in trust across the measured domains, and several mean scores decreased after use. These results should be interpreted as formative rather than as evidence of a controlled before-and-after effect, because the baseline survey reflected participants' general expectations and prior experiences with LLMs, whereas the post-use survey reflected their experience with the actual prototype. The scores may therefore have been influenced by several factors at once, including expectations about LLMs, answer quality, response time, terminology, interface presentation, varying levels of digital or AI familiarity, and the trust-supporting mechanisms themselves.

The qualitative findings provide a more nuanced interpretation. Participants recognized the value of source visibility, reliability information, and answer explanation, but also identified limitations in answer quality, source presentation, terminology, and interface design. These findings suggest that the trust-supporting mechanisms were considered relevant, but that their practical effect was constrained by the maturity and presentation of the proof-of-concept prototype. The group validation therefore supports the design direction of the framework, while also showing that improved answer quality, clearer source contribution, and better interface integration are needed before stronger claims about trust improvement can be made.

This finding reinforces the socio-technical nature of clinician trust. Trust does not emerge only from model performance, retrieval quality, or monitoring metrics. It also depends on how information is presented, how users understand the system, how the tool fits into clinical workflow, and whether professional responsibility remains clear. A trust-enabling framework can support better judgement, but it cannot guarantee appropriate use without training, clear role definitions, escalation procedures, interface refinement, and governance processes [20, 15, 17].

The framework also contributes to the operationalization of governance in clinical RAG systems. Existing literature and guidance emphasize auditability, lifecycle monitoring, post-deployment review, and accountability, but these principles can remain abstract unless they are translated into concrete system mechanisms [15, 17, 6]. In the Clinician Trust Framework, governance-oriented requirements are operationalized through trace logging, evaluation scores, feedback capture, warning generation, and monitoring endpoints. These mechanisms provide the raw material for future governance workflows, such as incident review, model or prompt updates, source corpus maintenance, threshold refinement, and escalation procedures. In this sense, the proof-of-concept demonstrates how governance can begin at the level of system architecture rather than being added only after deployment.

An important implication concerns the preservation of clinical reasoning and professional autonomy. The framework should not encourage clinicians to stop doing the work of interpretation themselves. Instead, it should support ongoing clinical judgement by encouraging source inspection, verification, questioning, and override. In this sense, the framework should not make clinicians dependent on AI-generated answers, but should help them inspect and challenge those answers more effectively. The goal is informed use rather than passive reliance.

5.1 Limitations

The current study has several limitations. First, although the validation included both a one-to-one expert session and a group validation with nurses, the sample remains limited. The group survey contained 16 baseline responses, 10 post-use responses, and 10 paired responses. This provides useful formative insight into usability, perceived transparency, and trust support, but it is not sufficient to generalize to nurses or clinicians more broadly. The one-to-one expert validation adds depth, but it should also be interpreted as expert feedback rather than as representative evidence.

Second, the group validation did not provide a controlled before-and-after comparison of the same RAG-based LLM assistant with and without the trust-supporting mechanisms. The baseline questionnaire reflected participants' general expectations and prior experiences with LLMs or AI systems, whereas the post-use questionnaire reflected their experience with the Clinician Trust Framework prototype. As a result, the paired survey results cannot be interpreted as direct evidence of the effect of the trust-supporting mechanisms alone. Differences between baseline and post-use responses may also reflect the specific prototype implementation, answer quality, response time, interface design, participant expectations, and varying levels of digital or AI familiarity. The quantitative findings should therefore be interpreted descriptively and formatively rather than as evidence of a direct causal effect.

Third, the validation was conducted in controlled evaluation settings outside routine clinical work. The use of realistic protocol-based scenarios made it possible to observe how participants interacted with the prototype, inspected sources, used feedback, and responded to monitoring information. However, the sessions do not yet show how the framework performs during everyday clinical workflow, where users may face time pressure, interruptions, incomplete questions, varying digital skills, different levels of AI familiarity, and competing care responsibilities.

Fourth, the user interface and answer presentation remain important limitations. The validation showed that participants valued source visibility and reliability information, but also found aspects of the output unclear or insufficiently structured. Some participants wanted faster access to the original protocol, clearer source contribution, more consistent language use, and better explanation of reliability terminology. This means that the framework's trust-supporting mechanisms cannot be evaluated separately from their interface presentation. If evalution scores, warnings, sources, or explanations are unclear, they may create confusion rather than support judgement.

Fifth, the proof-of-concept is not a production-ready hospital AI system. It demonstrates core mechanisms such as document ingestion, retrieval, answer generation, safety checks, evaluation scoring, warning generation, tracing, metrics, and feedback. However, several production requirements remain outside the current scope. These include hospital-grade authentication and authorization, role-based access control, immutable or tamper-evident audit logging, long-term data retention policies, formal incident management, fairness monitoring, drift detection, EHR integration, and validated clinical safety procedures. As a result, the framework should be understood as a reusable design foundation rather than as a deployable clinical decision-support system.

Sixth, security was not fully evaluated in the current proof-of-concept. This is an important limitation because the framework records prompts, retrieved sources, generated answers, traces, scores, and feedback. Future versions should therefore address authentication, authorization, secure trace storage, access governance for logs and dashboards, protection against prompt injection and jailbreak attempts, and prevention of data leakage through prompts or feedback. These controls are essential if such a framework is further developed for use in a hospital environment.

Seventh, the current evaluation scores and thresholds are still heuristic. They are useful for demonstrating how trust-relevant signals can be exposed, but they require further refinement before they can guide clinical decision-making or governance interventions. The current implementation should therefore be interpreted as a proof-of-concept demonstration of monitoring signals rather than as a clinically validated evaluation system.

Finally, the framework currently focuses on protocol-grounded question answering and does not evaluate downstream clinical outcomes. The framework can help clinicians inspect generated answers, but it does not demonstrate that use of the assistant improves care quality, reduces errors, saves time, or improves patient outcomes. Such claims would require a different evaluation design involving larger samples, clinical workflow studies, and possibly controlled comparisons.

5.2 Future Work

Future work should proceed in several directions. First, the evaluation should be expanded to include a larger and more diverse group of nurses and other healthcare professionals across multiple clinical roles.

Second, the scenario set should be broadened to include correct, partially correct, unsupported, ambiguous, and unsafe outputs. This would make it possible to assess whether the framework supports calibrated trust, including both appropriate reliance and appropriate scepticism.

Third, the user interface should be refined to ensure that citations, provenance information, warnings, and confidence or risk scores are understandable without overloading users. The validation suggests that source visibility is important, but that users also need clearer indication of how strongly each source contributed to the generated answer. Future iterations should therefore improve answer structure, source ranking, language consistency, reliability terminology, and access to the original protocol.

Fourth, the evaluation scores and thresholds used by the framework should be further refined and validated. More robust thresholds should be established through comparison with expert judgement, scenario-specific risk levels, and clinicians' interpretations of system behaviour. Future versions should also distinguish between signals intended for clinicians, developers, system administrators, and governance stakeholders. Not all monitoring information needs to be presented to all users in the same format or at the same level of detail. Some signals may be useful at the point of care, whereas others may be more appropriate for retrospective audit, dashboard-based monitoring, or technical debugging.

Fifth, future work should further develop the governance and security layer. This includes stronger audit controls, secure logging, drift detection, fairness monitoring, source freshness checks, access management, prompt-injection testing, and governance workflows for review and escalation. Deployment on hospital-approved infrastructure, such as an OpenShift-based environment, would also allow the framework to be evaluated under more realistic operational constraints.

Finally, future research should investigate how clinical reasoning can be preserved when clinicians use RAG-based LLM assistants. The goal should not be to make clinicians dependent on generated answers, but to support active interpretation, source inspection, verification, and challenge. This direction is important because trust calibration depends not only on system behaviour, but also on maintaining the professional skills needed to judge when AI support is appropriate.

6 Conclusion

This study investigated how clinician trust and transparency can be supported through the design and operationalization of a trust-enabling framework for clinical RAG-based LLM assistants interfacing with internal healthcare protocols. The resulting Clinician Trust Framework demonstrates how provenance, evaluation scores, warning generation, tracing, monitoring, refusal, clarification, and feedback mechanisms can be integrated around a RAG pipeline to make AI-generated protocol answers more inspectable and reviewable.

The main contribution of this study is not a clinical question-answering assistant, but a trust-enabling layer that turns generated answers into reviewable interactions. In the proof-of-concept, each answer is accompanied by supporting evidence, including retrieved sources, reliability-related signals, warnings, traces, metrics, and feedback. This provides a basis for clinical inspection, informed use, and future governance of RAG-based LLM assistants in protocol-driven clinical work.

The formative validation provides cautious but encouraging support for this design direction. The one-to-one expert validation indicated that the trust-supporting mechanisms can support informed answer assessment and trust calibration, while preserving the clinician's responsibility to verify and challenge outputs. The group validation produced more mixed quantitative results, but the qualitative findings showed that nurses recognized the relevance and potential value of source visibility, reliability information, answer explanation, and feedback. This suggests that the framework addresses a recognizable need: clinicians do not only need faster access to protocol information, but also support in judging whether AI-generated answers are grounded, appropriate, and safe to use.

The results should be interpreted in light of the formative validation setup and the maturity of the proof-of-concept prototype. The group validation was not designed as a controlled comparison between the same RAG-based assistant with and without the trust-supporting mechanisms, and the prototype still had limitations related to answer quality, response speed, terminology, source contribution, and interface presentation. These limitations likely influenced the quantitative trust scores. However, they do not undermine the overall direction of the work. Rather, they show that trust-supporting mechanisms must be developed together with stronger answer quality, clearer source presentation, better interface integration, and more robust evaluation thresholds.

Overall, this study shows that selected trust-supporting mechanisms can be operationalized in a working clinical RAG proof-of-concept and that clinical users recognized their potential value for evaluating AI-generated protocol answers. The project therefore provides a promising foundation for further development. Future work should focus on improving answer quality, response speed, source contribution, interface clarity, evaluation thresholds, and governance and security controls, followed by broader validation under more controlled and realistic clinical conditions. The goal is not to increase trust unconditionally, but to support calibrated trust: helping clinicians rely on AI-generated answers when they are well supported, and question, verify, or reject them when evidence is weak, uncertain, or unsafe.

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A Formative Validation Results: One-to-One Expert Validation

Evaluation & Validation Results

Enhancing Clinician Trust through a Trust-Enabling Framework for Clinical RAG-Based LLM Assistants

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This document evaluates the Clinician Trust Framework, a proof-of-concept architecture for making clinical RAG-based LLM answers more transparent, traceable, and reviewable. A scenario-based expert validation was conducted with a Head Research Nurse at Catharina Hospital Eindhoven. The participant first assessed an answer without monitoring information and then reassessed it with citations, provenance, metrics, warnings, explanations, dashboard signals, and audit/log information. After these mechanisms were shown, trust increased from 3/5 to 5/5 and decision confidence from 1/5 to 5/5. The participant shifted from uncertainty and possible rejection toward acceptance with verification, indicating improved trust calibration rather than blind trust. Although limited to one expert participant, the evaluation provides formative evidence that the framework supports clinicians in judging, verifying, and appropriately relying on AI-generated answers.

1. Introduction

This document presents the evaluation and validation results of the Clinician Trust Framework, a proof-of-concept architecture designed to support clinician trust in clinical RAG-based LLM assistants. The evaluation focused on whether the framework helps users better understand, inspect, and judge AI-generated answers by exposing trust-supporting information such as citations, provenance, evaluation metrics, warnings, explanations, dashboard signals, and audit/log information.

The evaluation assumes that trust is not only a subjective feeling, but a measurable socio-technical outcome. Therefore, the evaluation assessed not only perceived trust, but also decision confidence, behavioural reliance, source inspection, and trust calibration. This aligns with the evaluation plan, which defines trust through transparency/explainability, traceability/auditability, reliability/safety, human control/autonomy, and trust calibration.

The goal of the evaluation was not to prove that the system is ready for clinical deployment. Instead, the purpose was to validate whether the implemented trust-enabling mechanisms can make a clinical RAG interaction more transparent, traceable, evaluable, and open to human review. In this context, the evaluation should be interpreted as a formative validation of the framework's design direction and practical relevance.

2. Methodology

The evaluation used a scenario-based expert validation approach. In this design, the participant first evaluated a system answer without additional monitoring information. After that, the same answer was evaluated again with trust-supporting information available, such as citations, metrics, warnings, logs, and explanations. This setup made it possible to compare how monitoring information influenced trust perception, decision-making behaviour, and trust calibration. The evaluation plan uses this before-and-after structure to assess the relationship between system mechanisms, user perception, user behaviour, and trust calibration.

The evaluation instrument consisted of three main parts. First, a pre-study section captured participant background and baseline attitudes toward AI. Second, a scenario-based form measured trust before and after monitoring information was shown, including decision confidence, intended action, source inspection, and ratings for trust mechanisms. Third, a post-study reflection captured the participant's overall judgement of the system and the perceived usefulness of individual mechanisms.

The session was conducted with one expert participant, a Head Research Nurse in the geriatrics department at Catharina Hospital Eindhoven (CZE). The participant has 23 years of nursing experience at CZE, and research experience in the field of AI. The participant also uses AI/LLM tools daily, mainly for work-related support and code generation.

This participant was selected as an expert key informant because of their combined clinical, organizational, and AI-related perspective. Due to practical constraints, it was difficult to organize a broader evaluation with multiple nurses within the available project period. Instead, the evaluation used an expert-informed validation approach, in which a clinically experienced and AI-aware nurse researcher assessed whether the framework's trust-supporting mechanisms were understandable, useful, and relevant for nursing practice.

The results should therefore be interpreted as formative expert validation rather than statistically generalizable evidence across all nurses. The evaluation provides a grounded indication of whether the framework aligns with clinical nursing concerns, while broader validation with more participants remains necessary in future work.

3. Results

3.1. Participant baseline attitude

The participant showed a balanced baseline attitude toward AI systems. General trust in AI was moderate, while confidence in evaluating AI-generated outputs and preference for explanations were high.

Baseline item	Score
Trust in AI systems in general	3/5
Trust in AI to support clinical decision-making	3/5
Confidence in evaluating AI-generated outputs	5/5
Preference for explanations alongside AI outputs	5/5
Attitude toward AI use in a clinical context	3/5

The participant indicated that using AI can be acceptable when users understand the model's capabilities and shortcomings and actively check the answers. However, an important concern was whether all users would verify LLM outputs in practice. This highlights that trust is not only related to system quality, but also to user behaviour, professional responsibility, workflow integration, and oversight.

3.2. Trust and decision confidence before and after mechanisms

The evaluation showed a clear increase in trust and decision confidence after the trust mechanisms were made available.

Measure	Before monitoring	After monitoring	Change
Trust in the answer	3/5	5/5	+2
Confidence in decision/action	1/5	5/5	+4

Before monitoring information was shown, the participant gave the answer a moderate trust score of 3/5, but decision confidence was very low at 1/5. This suggests that a correct answer alone was not sufficient to support confident clinical decision-making. The participant selected verify further, reject, and escalate as possible actions, indicating uncertainty about how to interpret or rely on the answer.

After monitoring information was shown, trust increased to 5/5 and decision confidence increased to 5/5. The final decision changed to accept while still selecting verify. This is an important result because it shows that the monitoring information did not lead to blind trust. Instead, it supported more informed trust, the participant became more confident in the answer, while still maintaining a verification-oriented clinical attitude.

3.3. Source inspection behaviour

The participant's source inspection behaviour changed after monitoring information was shown.

Moment	Source inspection
Before monitoring	Thorough
After monitoring	Brief

Before monitoring, the participant indicated a need for thorough source inspection. After monitoring, source inspection changed to brief. This suggests that the additional system information made the answer easier to evaluate, because citations, provenance, metrics, explanations, and warnings provided faster insight into the reliability and origin of the answer.

However, the participant also noted that it can be ambiguous when an answer is shown in the user interface, making it difficult to interpret. This indicates a usability issue, trust-supporting mechanisms are valuable, but they must be presented clearly and in a way that does not create confusion.

3.4. Trust mechanisms ratings

The ratings for individual trust mechanisms improved strongly after monitoring information was shown.

Trust mechanism	Item	Before	After	Change
Transparency/Explainability	The system clearly explains how the answer was generated	2	5	+3
Transparency/Explainability	I understand why this answer was produced	1	5	+4
Traceability/Auditability	I can trace the origin of the information	3	5	+2
Traceability/Auditability	I could reconstruct how the system arrived at this answer	1	5	+4
Reliability/Safety	The answer appears clinically reliable	5	5	0
Reliability/Safety	The system signals help assess risk	1	5	+4
Reliability/Safety	The system behaviour is appropriate	2	5	+3
Human Control/Autonomy	I feel able to challenge or	1	5	+4

	override the system			
Human Control/Autonomy	The system supports decision-making rather than replacing it	3	5	+2
Usefulness	The additional system information is useful	1	5	+4
Usefulness	The system behaviour is useful	2	5	+3
Usefulness	The system improves my ability to make a decision	2	5	+3

The largest improvements were found in understanding why the answer was produced, reconstructing how the system arrived at the answer, assessing risk using system signals, feeling able to challenge or override the system, and judging the usefulness of additional system information. These findings support the assumption that trust-supporting mechanisms can improve the user's ability to evaluate and interpret AI-generated answers.

3.5. Behavioural trust and final decision

Before monitoring information was shown, the participant did not simply accept the answer. The selected actions were:

- Verify further
- Reject
- Escalate

After monitoring information was shown, the selected actions changed to:

- Accept
- Verify

This indicates a shift from uncertainty and possible rejection toward acceptance with verification. In other words, the monitoring information helped the participant move from low decision confidence to a more calibrated decision. The answer was accepted because it became easier to inspect and justify, but verification remained part of the clinical decision-making behaviour.

This is a positive outcome for the framework, because the objective is not to maximize trust uncritically. The objective is to support appropriate reliance. In this case, the participant's final behaviour suggests aligned trust, a correct answer was accepted but not treated as a replacement for professional judgement.

3.6. Post-study survey

The post-study survey showed strong overall agreement that the system improved the participant's ability to evaluate AI-generated answers.

Post-study item	Score
The system improved my ability to evaluate answers	5/5
Monitoring information increased my trust appropriately	5/5
The system helped me detect incorrect answers	5/5
The system reduced my confidence in unreliable outputs	5/5
I would use this system in clinical practice	5/5
The system helped me avoid overtrust	5/5
The system helped me avoid undertrust	5/5
I felt my trust aligned with answer quality	5/5

These results suggest that the participant perceived the trust mechanisms as useful for trust calibration. The participant did not only report increased trust but also indicated that the system could help reduce confidence in unreliable outputs. This distinction is important because a trustworthy clinical AI assistant should support both justified trust and justified distrust.

3.7. Mechanism Usefulness

The participant ranked the most useful mechanisms as follows:

Mechanism	Ranking
Citations/provenance	1
Metrics (eg. groundedness)	1
Alerts/warnings	1
Dashboard	2
Explanation text/follow-up questions	2
Audit/log information	3

The highest-ranked mechanisms were citations/provenance, metrics, and alerts/warnings. This indicates that the participant valued information that directly helps determine whether an answer is grounded, reliable, and safe to interpret. Dashboard information and explanation/follow-up questions were also considered valuable, while audit/log information was useful but ranked slightly lower, likely because it is more relevant for retrospective review than immediate decision-making.

4. Discussion

The evaluation results indicate that the Clinician Trust Framework can support more informed clinician trust by making LLM generated answers more transparent, traceable, and evaluable. The most important finding is that the participant's trust and decision confidence increased substantially after monitoring information was shown. Trust increased from 3/5 to 5/5, while decision confidence increased from 1/5 to 5/5.

This suggests that the answer itself was not enough to support confident use. Even when the answer was correct, the participant initially lacked sufficient confidence to act on it. Once the answer was accompanied by citations, provenance, metrics, warnings, explanations, dashboard signals, and audit/log information, the participant was better able to judge the answer's reliability and limitations.

The results also show that the framework supported trust calibration rather than blind trust. After monitoring information was shown, the participant accepted the answer but still selected verification as part of the final decision. This is an important distinction. The system did not remove the need for professional judgement instead, it made the answer more inspectable and easier to evaluate.

The qualitative feedback reinforces this interpretation. The participant stated that it is acceptable to work with AI when users understand the model's capabilities and shortcomings and check the answers. However, the participant also expressed concern about whether all users would verify LLM outputs in practice. This supports the broader argument that clinician trust is not only a technical issue, but also a socio-technical issue involving user behaviour, workflow design, oversight, and accountability.

A limitation of the evaluation is that it was conducted with one participant. However, the participant was not selected as a convenience user only, but as an expert key informant. Their role as Head Research Nurse in the geriatrics department, combined with 23 years of clinical nursing experience at Catharina Hospital Eindhoven and AI-related research experience, makes their feedback especially valuable for formative validation. The participant could evaluate the framework from multiple perspectives: as a nurse, as a clinical researcher, as an AI-aware user, and as someone familiar with the practical concerns of other nurses.

For this reason, the evaluation should be understood as an expert validation session that provides depth rather than breadth. It does not replace broader validation with a larger and more diverse group of nurses, but it does provide a meaningful first assessment of whether the trust mechanisms are clinically understandable, practically useful, and aligned with nursing concerns.

Another limitation is that the user interface may introduce ambiguity. The participant noted that the way an answer is presented in the UI can make interpretation difficult. This suggests that future work should not only improve the underlying monitoring mechanisms, but also focus on how those mechanisms are communicated to clinicians. Metrics, warnings, sources, and explanations must be understandable at the point of use, otherwise they may create confusion instead of trust.

Future evaluation should include more participants, multiple clinical roles, and a broader set of scenarios. This would allow stronger assessment of whether the framework reduces overtrust in incorrect outputs and encourages appropriate reliance on correct outputs.

Overall, the evaluation provides initial evidence that the framework supports the intended trust-enabling function. By turning a generated answer into a reviewable interaction with sources, metrics, warnings, explanations, traces, and feedback, the system helps clinicians better understand, verify, and calibrate their reliance on AI-generated outputs.

B Formative Validation Results: Group Validation with Nurses

Comparing Baseline and Follow-Up Survey Results for the Clinician Trust Framework

This notebook compares responses from two survey files:

- `survey_before.xlsx` : baseline survey responses collected before use of the trust framework, based on participants' existing experience and expectations of LLM-based tools.
- `survey_after.xlsx` : follow-up survey responses collected after clinicians used the implemented trust framework during the validation session.

The purpose of this analysis is to explore whether the implemented trust framework is associated with changes in clinicians' trust in RAG-based large language model (LLM) generated answers. The notebook combines quantitative survey interpretation with the qualitative validation picture from the group-session notes and open survey responses. The goal is not to prove clinical effectiveness, but to identify whether the framework mechanisms appear relevant, useful, understandable, and in need of refinement.

In [1]:

```
import re
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

pd.set_option("display.max_columns", 120)
pd.set_option("display.max_colwidth", 120)
pd.options.display.float_format = "{:.6f}".format

DATA_DIR = Path("data")
if not (DATA_DIR / "survey_before.xlsx").exists():
    DATA_DIR = Path(".")

before_path = DATA_DIR / "survey_before.xlsx"
after_path = DATA_DIR / "survey_after.xlsx"

before_raw = pd.read_excel(before_path)
```

```
after_raw = pd.read_excel(after_path)

print("Before shape:", before_raw.shape)
print("After shape:", after_raw.shape)
```

```
Before shape: (16, 42)
After shape: (10, 48)
```

The loaded files contain 16 baseline rows and 10 follow-up rows. This already shows that the before and after datasets are not equal in size. For that reason, comparisons that use all responses are useful as general descriptive context, but they should not be treated as the strongest evidence of change. The more important analysis is the paired comparison, where the same participant can be followed from baseline to follow-up.

The baseline survey contains 16 responses, while the follow-up survey contains 10 responses. After cleaning the participant identifiers, 10 respondents can be matched across the baseline and follow-up surveys. This means the paired analysis is more informative than a simple overall before/after comparison, because it compares the same respondents before and after using the prototype.

The analysis should still be interpreted as formative and descriptive. The sample is small, the validation took place in a prototype setting, and the results are affected by practical implementation limitations such as answer generation quality, response time, language consistency, and UI clarity, as well as the limited participant technical expertise.

1. Clean Column Names and Participant Identifiers

The survey files use Dutch question text as column names, and some columns include non-breaking spaces. The code below standardizes the column names and cleans the `u-nummer` field so that responses can be anonymously paired across the baseline and post-use surveys.

In [2]:
```
def normalize_text(value):
    # Normalize text values while preserving meaning.
    if pd.isna(value):
        return np.nan
    text = str(value).replace("\xa0", " ").strip()
    text = re.sub(r"\s+", " ", text)
    return text
```

```
def normalize_column_name(col):
    return normalize_text(col).lower().rstrip()


def make_unique_columns(cols):
    # Ensure normalized column names remain unique.
    seen = {}
    out = []
    for col in cols:
        base = col
        if base not in seen:
            seen[base] = 0
            out.append(base)
        else:
            seen[base] += 1
            out.append(f"{base}__{seen[base]}")
    return out


def clean_participant_id(value):
    # Normalize u-numbers such as 'uXXXXXX', 'UXXXXXX', or 'XXXXXX' to a common key.
    if pd.isna(value):
        return np.nan
    text = str(value).strip().lower().replace("u", "")
    digits = re.sub(r"\D", "", text)
    return digits if digits else np.nan


def clean_frame(df):
    df = df.copy()
    df.columns = make_unique_columns([normalize_column_name(c) for c in df.columns])
    id_col = next(c for c in df.columns if "u-nummer" in c)
    df["participant_key"] = df[id_col].map(clean_participant_id)
    return df

before = clean_frame(before_raw)
after = clean_frame(after_raw)

paired_ids = sorted(set(before["participant_key"].dropna()).intersection(after["participant_key"].dropna()))

print(f"Before responses: {len(before)}")
```

```python
print(f"After responses:  {len(after)}")
print(f"Paired responses: {len(paired_ids)}")
```

```
Before responses: 16
After responses:  10
Paired responses: 10
```

After standardizing the participant identifiers, 10 paired responses are available. This is the most important sample for assessing before/after change, because each participant acts as their own baseline. The remaining baseline-only responses can still provide context, but they should not drive conclusions about change caused by using the prototype.

In [3]:
```python
LIKERT_MAP = {
    # Agreement scale
    "helemaal oneens": 1,
    "oneens": 2,
    "niet eens, niet oneens": 3,
    "neutraal": 3,
    "eens": 4,
    "helemaal eens": 5,

    # Trust scale variants
    "helemaal geen vertrouwen": 1,
    "geen vertrouwen": 1,
    "weinig vertrouwen": 2,
    "vertrouwen": 4,
    "veel vertrouwen": 5,
    "heel veel vertrouwen": 5,

    # Certainty scale variants
    "heel onzeker": 1,
    "onzeker": 2,
    "zeker": 4,
    "heel zeker": 5,
    "zeer zeker": 5,
}


def to_likert_score(value):
    if pd.isna(value):
        return np.nan
```

```
    text = normalize_text(value).lower().rstrip(".")
    return LIKERT_MAP.get(text, np.nan)


def score_columns(df, columns):
    return df[columns].apply(lambda col: col.map(to_likert_score))

# Quick check of response values that were successfully mapped.
for label, df in {"before": before, "after": after}.items():
    mapped_values = []
    for col in df.columns:
        scores = df[col].map(to_likert_score)
        if scores.notna().any():
            mapped_values.extend(df.loc[scores.notna(), col].astype(str).str.lower().unique().tolist())
    print(label, "mapped answer variants:", sorted(set(mapped_values)))
```

```
before mapped answer variants: ['eens', 'heel zeker', 'helemaal eens', 'helemaal oneens', 'neutraal', 'niet eens, ni
et oneens', 'oneens', 'vertrouwen', 'weinig vertrouwen', 'zeker']
after mapped answer variants: ['eens', 'geen vertrouwen', 'heel onzeker', 'helemaal eens', 'helemaal oneens', 'neutr
aal', 'niet eens, niet oneens', 'niet eens, niet oneens.', 'oneens', 'onzeker', 'vertrouwen', 'weinig vertrouwen',
'zeker']
```

The Likert response categories were mapped to numeric values from 1 to 5. This makes it possible to calculate averages and differences, but the results should still be read carefully because Likert answers are ordinal survey responses rather than precise continuous measurements. A difference of `+0.4`, for example, should be interpreted as a descriptive shift in response tendency, not as an exact measurement of trust.

In [4]:
```
META_PATTERNS = [
    "id", "start time", "completion time", "email", "name", "last modified", "u-nummer",
    "opmerkingen", "consent", "toestemming", "akkoord", "leeftijd", "afdeling", "rol",
    "ai tool", "ervaring", "huidige", "eventuele", "informed", "schermopnames"
]


def is_metadata_column(col):
    return any(pattern in col for pattern in META_PATTERNS)

common_cols = sorted(set(before.columns).intersection(after.columns))
comparable_questions = []
```

```
for col in common_cols:
    if is_metadata_column(col):
        continue
    before_scores = before[col].map(to_likert_score)
    after_scores = after[col].map(to_likert_score)
    if before_scores.notna().any() and after_scores.notna().any():
        comparable_questions.append(col)

print(f"Comparable Likert questions: {len(comparable_questions)}")
for i, q in enumerate(comparable_questions, 1):
    print(f"{i:02d}. {q}")
```

```
Comparable Likert questions: 12
01. de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.
02. de indeling van het antwoord helpt me om het risico van het gebruik van het antwoord in te schatten.
03. het antwoord lijkt me klinisch betrouwbaar.
04. het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.
05. hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u de monitoring informatie hebt ge
zien?
06. hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie heeft gezien?
07. ik begrijp hoe het antwoord tot stand is gekomen.
08. ik begrijp waar de informatie in het antwoord vandaan komt.
09. ik kan ontdekken hoe het model tot dit antwoord is gekomen.
10. ik kan tegen het systeem ingaan.
11. ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.
12. ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn eigen kunde (klinisch redeneren).
```

There are 12 overlapping Likert questions that can be compared between the baseline and follow-up surveys. These questions form the basis for the before/after comparison. Questions that were only asked after the prototype session are analysed separately later, because they cannot be compared directly with baseline answers.

In [5]:
```
def shorten(text, max_len=80):
    return text if len(text) <= max_len else text[: max_len - 1] + "…"


def question_level_summary(before, after, questions):
    rows = []
    for q in questions:
        b = before[q].map(to_likert_score)
        a = after[q].map(to_likert_score)
```

```
        rows.append({
            "question": q,
            "short_question": shorten(q),
            "before_n": int(b.notna().sum()),
            "before_mean": b.mean(),
            "before_median": b.median(),
            "after_n": int(a.notna().sum()),
            "after_mean": a.mean(),
            "after_median": a.median(),
            "mean_diff_after_minus_before": a.mean() - b.mean(),
        })
    return pd.DataFrame(rows)

question_summary = question_level_summary(before, after, comparable_questions)
question_summary.sort_values("mean_diff_after_minus_before", ascending=False)
```

Out[5]:

	question	short_question	before_n	before_mean	before_median	after_n	after_mean	after_median	mean_diff_after_minu
10	ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.	ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.	16	3.250000	3.000000	10	3.700000	4.000000	
3	het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.	het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.	16	3.375000	3.000000	10	3.400000	3.000000	
11	ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn eigen kunde (klinisch redeneren).	ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn ei...	16	3.687500	4.000000	10	3.700000	4.000000	
7	ik begrijp waar de informatie in het antwoord vandaan komt.	ik begrijp waar de informatie in het antwoord vandaan komt.	16	3.437500	4.000000	10	3.400000	3.500000	-
1	de indeling van het antwoord helpt me om het risico	de indeling van het antwoord helpt me om het risico van het	16	3.250000	3.000000	10	3.000000	3.000000	-

	question	short_question	before_n	before_mean	before_median	after_n	after_mean	after_median	mean_diff_after_minu
	van het gebruik van het antwoord in te schatten.	gebruik van het ant...							
8	ik kan ontdekken hoe het model tot dit antwoord is gekomen.	ik kan ontdekken hoe het model tot dit antwoord is gekomen.	16	3.562500	4.000000	10	3.300000	3.000000	-
6	ik begrijp hoe het antwoord tot stand is gekomen.	ik begrijp hoe het antwoord tot stand is gekomen.	16	3.437500	3.500000	10	3.100000	3.000000	-
0	de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.	de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.	16	3.687500	4.000000	10	3.200000	3.000000	-
5	hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie heeft gezien?	hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie hee...	16	3.125000	3.000000	10	2.600000	3.000000	-
2	het antwoord lijkt me	het antwoord lijkt me klinisch betrouwbaar.	16	3.062500	3.000000	10	2.500000	3.000000	-

	question	short_question	before_n	before_mean	before_median	after_n	after_mean	after_median	mean_diff_after_minu
	klinisch betrouwbaar.								
9	ik kan tegen het systeem ingaan.	ik kan tegen het systeem ingaan.	16	3.687500	4.000000	10	2.900000	3.000000	-
4	hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u de monitoring informatie hebt gezien?	hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u...	16	3.375000	3.000000	10	2.500000	3.000000	-

This table compares all available baseline responses with all available follow-up responses at question level. It gives a broad overview, but it is not the cleanest measure of change because the baseline group contains more responses than the follow-up group.

The overall pattern is mixed and slightly negative. The clearest positive result is source visibility: respondents scored higher after the session on being able to see which sources were used to generate the answer. However, several items decreased, especially clinical reliability, certainty after seeing monitoring information, trust after monitoring information, and the feeling of being able to go against the system. This suggests that the current prototype did not simply increase trust. It made some trust-relevant information more visible, but participants were not yet more confident in the generated answers overall.

In [6]:
```
def paired_question_summary(before, after, questions, paired_ids):
    rows = []
    b_idx = before.set_index("participant_key")
    a_idx = after.set_index("participant_key")

    for q in questions:
        b = b_idx.loc[paired_ids, q].map(to_likert_score)
        a = a_idx.loc[paired_ids, q].map(to_likert_score)
        diff = a - b
```

```
        rows.append({
            "question": q,
            "short_question": shorten(q),
            "paired_n": int(diff.notna().sum()),
            "paired_before_mean": b.mean(),
            "paired_after_mean": a.mean(),
            "paired_mean_diff": diff.mean(),
            "n_increased": int((diff > 0).sum()),
            "n_same": int((diff == 0).sum()),
            "n_decreased": int((diff < 0).sum()),
        })
    return pd.DataFrame(rows)

paired_question_results = paired_question_summary(before, after, comparable_questions, paired_ids)
paired_question_results.sort_values("paired_mean_diff", ascending=False)
```

Out[6]:

	question	short_question	paired_n	paired_before_mean	paired_after_mean	paired_mean_diff	n_increased	n_same	n_de
10	ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.	ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.	10	3.300000	3.700000	0.400000	5	4	
3	het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.	het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.	10	3.300000	3.400000	0.100000	2	7	
7	ik begrijp waar de informatie in het antwoord vandaan komt.	ik begrijp waar de informatie in het antwoord vandaan komt.	10	3.300000	3.400000	0.100000	1	8	
11	ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn eigen kunde (klinisch redeneren).	ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn ei...	10	3.600000	3.700000	0.100000	2	7	
1	de indeling van het antwoord helpt me om het risico	de indeling van het antwoord helpt me om het risico van het	10	3.200000	3.000000	-0.200000	1	7	

	question	short_question	paired_n	paired_before_mean	paired_after_mean	paired_mean_diff	n_increased	n_same	n_de
	van het gebruik van het antwoord in te schatten.	gebruik van het ant...							
6	ik begrijp hoe het antwoord tot stand is gekomen.	ik begrijp hoe het antwoord tot stand is gekomen.	10	3.400000	3.100000	-0.300000	2	4	
8	ik kan ontdekken hoe het model tot dit antwoord is gekomen.	ik kan ontdekken hoe het model tot dit antwoord is gekomen.	10	3.600000	3.300000	-0.300000	1	5	
0	de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.	de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.	10	3.600000	3.200000	-0.400000	1	4	
5	hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie heeft gezien?	hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie hee...	10	3.100000	2.600000	-0.500000	0	6	
9	ik kan tegen het systeem ingaan.	ik kan tegen het systeem ingaan.	10	3.500000	2.900000	-0.600000	0	5	

	question	short_question	paired_n	paired_before_mean	paired_after_mean	paired_mean_diff	n_increased	n_same	n_de
2	het antwoord lijkt me klinisch betrouwbaar.	het antwoord lijkt me klinisch betrouwbaar.	10	3.200000	2.500000	-0.700000	1	4	
4	hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u de monitoring informatie hebt gezien?	hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u...	10	3.500000	2.500000	-1.000000	0	4	

This paired table is more important than the previous unpaired table because it compares the same 10 respondents before and after the validation session. The pattern again does not show a broad increase in trust.

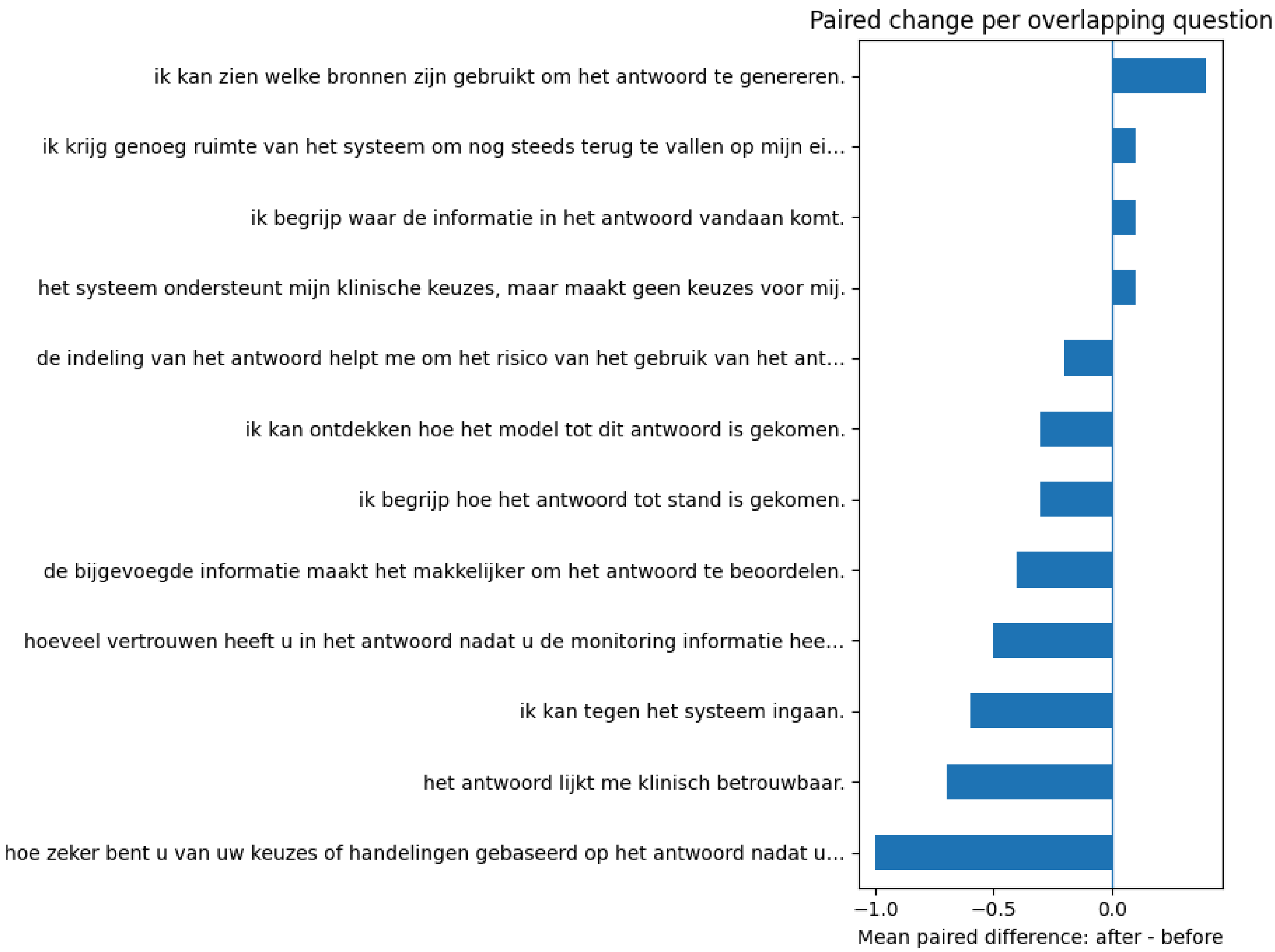
The strongest positive paired change is that participants reported being better able to see which sources were used to generate the answer. This supports the relevance of the citation and source-visibility mechanism. However, the largest negative changes are related to certainty and trust after seeing reliability information, as well as perceived clinical reliability. This indicates that the trust information did not automatically increase confidence. In a clinical context, that is not necessarily a failure: lower certainty may reflect more critical assessment. However, the qualitative notes suggest that answer quality, answer structure, and unclear presentation also contributed to these lower ratings.

In [7]:
```
# Visualize paired question differences.
# Positive values mean the after score was higher than the before score for matched respondents.
plot_df = paired_question_results.sort_values("paired_mean_diff")

ax = plot_df.plot.barh(
    x="short_question",
    y="paired_mean_diff",
    figsize=(9, 7),
```

```
    legend=False,
)
ax.axvline(0, linewidth=1)
ax.set_xlabel("Mean paired difference: after − before")
ax.set_ylabel("")
ax.set_title("Paired change per overlapping question")
plt.tight_layout()
plt.show()

# save image in output directory
OUTPUT_DIR = DATA_DIR / "survey_analysis_outputs"
OUTPUT_DIR.mkdir(exist_ok=True)
ax.get_figure().savefig(OUTPUT_DIR / "paired_question_differences.png", bbox_inches="tight")
```


Paired change per overlapping question
ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.
ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn ei...
ik begrijp waar de informatie in het antwoord vandaan komt.
het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.
de indeling van het antwoord helpt me om het risico van het gebruik van het ant...
ik kan ontdekken hoe het model tot dit antwoord is gekomen.
ik begrijp hoe het antwoord tot stand is gekomen.
de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.
hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie hee...
ik kan tegen het systeem ingaan.
het antwoord lijkt me klinisch betrouwbaar.
hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u...
−1.0
−0.5
0.0
Mean paired difference: after - before

The figure makes the question-level pattern easier to see. Most bars are close to zero or slightly below zero, which means the follow-up ratings were mostly similar to or slightly lower than the baseline ratings. The main positive exception is source visibility.

This visual supports a cautious conclusion: the prototype did not produce a clear positive trust effect across all questions, but it did appear to improve the visibility of sources. The negative bars should be interpreted together with the qualitative data, because participants were reacting not only to the trust framework concept, but also to the quality of the answers, speed, language, and layout of the prototype implementation.

In [8]:
```
DOMAIN_MAP = {
    "Answer transparency": [
        "het systeem legt duidelijk uit hoe het antwoord is gegenereerd.",
        "ik begrijp hoe het antwoord tot stand is gekomen.",
        "de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.",
    ],
    "Provenance / traceability": [
        "ik begrijp waar de informatie in het antwoord vandaan komt.",
        "ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.",
        "ik kan ontdekken hoe het model tot dit antwoord is gekomen.",
    ],
    "Clinical reliability / risk visibility": [
        "het antwoord lijkt me klinisch betrouwbaar.",
        "het systeem gedraagt zich gepast voor een klinisch ondersteunende ai tool.",
        "de indeling van het antwoord helpt me om het risico van het gebruik van het antwoord in te schatten.",
        "het systeem maakt (on)zekerheid en limitaties zichtbaar wanneer deze er is/zijn.",
    ],
    "Human oversight / autonomy": [
        "ik kan tegen het systeem ingaan.",
        "het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.",
        "ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn eigen kunde (klinisch redener
    ],
    "Trust/certainty after monitoring": [
        "hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie heeft gezien?",
        "hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u de monitoring informatie h
    ],
}

# Keep only items that actually exist in both files.
DOMAIN_MAP = {
```

```
    domain: [q for q in questions if q in before.columns and q in after.columns]
    for domain, questions in DOMAIN_MAP.items()
}

DOMAIN_MAP
```

Out[8]:

```
{'Answer transparency': ['het systeem legt duidelijk uit hoe het antwoord is gegenereerd.',
  'ik begrijp hoe het antwoord tot stand is gekomen.',
  'de bijgevoegde informatie maakt het makkelijker om het antwoord te beoordelen.'],
 'Provenance / traceability': ['ik begrijp waar de informatie in het antwoord vandaan komt.',
  'ik kan zien welke bronnen zijn gebruikt om het antwoord te genereren.',
  'ik kan ontdekken hoe het model tot dit antwoord is gekomen.'],
 'Clinical reliability / risk visibility': ['het antwoord lijkt me klinisch betrouwbaar.',
  'het systeem gedraagt zich gepast voor een klinisch ondersteunende ai tool.',
  'de indeling van het antwoord helpt me om het risico van het gebruik van het antwoord in te schatten.',
  'het systeem maakt (on)zekerheid en limitaties zichtbaar wanneer deze er is/zijn.'],
 'Human oversight / autonomy': ['ik kan tegen het systeem ingaan.',
  'het systeem ondersteunt mijn klinische keuzes, maar maakt geen keuzes voor mij.',
  'ik krijg genoeg ruimte van het systeem om nog steeds terug te vallen op mijn eigen kunde (klinisch redenere
n).'],
 'Trust/certainty after monitoring': ['hoeveel vertrouwen heeft u in het antwoord nadat u de monitoring informatie
heeft gezien?',
  'hoe zeker bent u van uw keuzes of handelingen gebaseerd op het antwoord nadat u de monitoring informatie hebt g
ezien?']}
```

The individual questions are grouped into broader trust-framework domains. This helps connect the survey results to the conceptual purpose of the framework: answer transparency, provenance and traceability, clinical reliability and risk visibility, human oversight and autonomy, and trust/certainty after monitoring.

These domain scores are useful for summarizing the results, but they should not hide the underlying nuance. For example, provenance and traceability may improve because sources are visible, while trust and certainty may decrease because the answer itself is not yet reliable or clearly presented.

In [9]:

```
def add_domain_scores(df, domain_map):
    df = df.copy()
    for domain, questions in domain_map.items():
        if not questions:
            df[f"domain::{domain}"] = np.nan
        else:
```

```
            df[f"domain::{domain}"] = score_columns(df, questions).mean(axis=1)
    return df

before_domains = add_domain_scores(before, DOMAIN_MAP)
after_domains = add_domain_scores(after, DOMAIN_MAP)

domain_cols = [f"domain::{domain}" for domain in DOMAIN_MAP]

domain_unpaired = pd.DataFrame([
    {
        "domain": col.replace("domain::", ""),
        "before_n": int(before_domains[col].notna().sum()),
        "before_mean": before_domains[col].mean(),
        "after_n": int(after_domains[col].notna().sum()),
        "after_mean": after_domains[col].mean(),
        "mean_diff_after_minus_before": after_domains[col].mean() - before_domains[col].mean(),
    }
    for col in domain_cols
])

domain_unpaired.sort_values("mean_diff_after_minus_before", ascending=False)
```

Out[9]:

	domain	before_n	before_mean	after_n	after_mean	mean_diff_after_minus_before
1	Provenance / traceability	16	3.416667	10	3.466667	0.050000
3	Human oversight / autonomy	16	3.583333	10	3.333333	-0.250000
2	Clinical reliability / risk visibility	16	3.140625	10	2.850000	-0.290625
0	Answer transparency	16	3.416667	10	3.100000	-0.316667
4	Trust/certainty after monitoring	16	3.250000	10	2.550000	-0.700000

The unpaired domain comparison shows that provenance and traceability is the only domain with a small positive average change. This is consistent with participants noticing and valuing source-related information. The other domains show negative average changes, with the largest decrease in trust/certainty after monitoring.

Because this table uses all available responses rather than only paired respondents, it should be treated as descriptive context. The pattern suggests that the trust mechanisms were not enough to overcome concerns about the current prototype, especially answer quality and clarity.

In [10]:
```
def paired_domain_summary(before_domains, after_domains, domain_cols, paired_ids):
    b_idx = before_domains.set_index("participant_key")
    a_idx = after_domains.set_index("participant_key")
    rows = []
    for col in domain_cols:
        b = b_idx.loc[paired_ids, col]
        a = a_idx.loc[paired_ids, col]
        diff = a - b
        rows.append({
            "domain": col.replace("domain::", ""),
            "paired_n": int(diff.notna().sum()),
            "paired_before_mean": b.mean(),
            "paired_after_mean": a.mean(),
            "paired_mean_diff": diff.mean(),
            "n_increased": int((diff > 0).sum()),
            "n_same": int((diff == 0).sum()),
            "n_decreased": int((diff < 0).sum()),
        })
    return pd.DataFrame(rows)

domain_paired = paired_domain_summary(before_domains, after_domains, domain_cols, paired_ids)
domain_paired.sort_values("paired_mean_diff", ascending=False)
```

Out[10]:

	domain	paired_n	paired_before_mean	paired_after_mean	paired_mean_diff	n_increased	n_same	n_decreased
1	Provenance / traceability	10	3.400000	3.466667	0.066667	3	5	2
3	Human oversight / autonomy	10	3.466667	3.333333	-0.133333	3	2	5
0	Answer transparency	10	3.400000	3.100000	-0.300000	2	1	7
2	Clinical reliability / risk visibility	10	3.175000	2.850000	-0.325000	2	2	6
4	Trust/certainty after monitoring	10	3.300000	2.550000	-0.750000	0	4	6

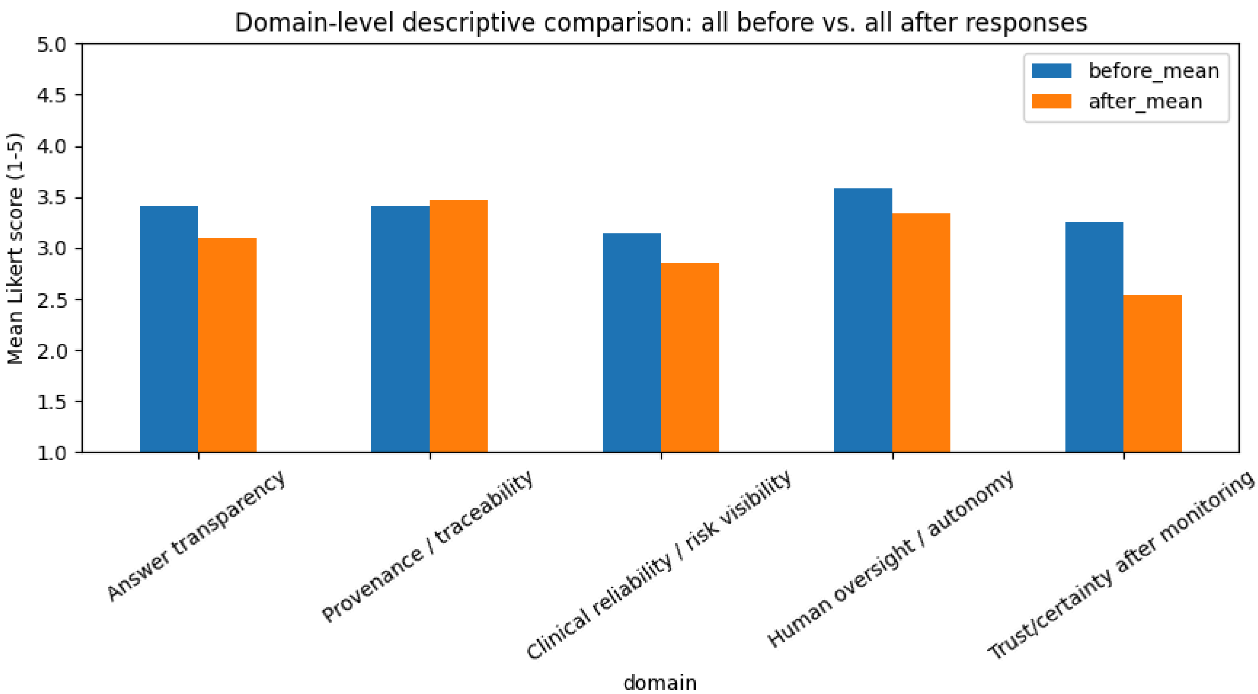
The paired domain comparison gives the clearest quantitative picture. Among the same 10 respondents, provenance and traceability increased slightly, while answer transparency, clinical reliability/risk visibility, human oversight/autonomy, and trust/certainty after trust indicators decreased.

The largest decrease is in trust/certainty after reliability information. Importantly, this should not be interpreted only as a negative outcome. A trust framework should not blindly increase trust, it should support calibrated trust. If reliability information makes users more cautious, that can be desirable. However, the qualitative findings indicate that participants were also influenced by prototype limitations, such as incorrect or unclear generated answers, slow responses, mixed Dutch/English output, and unclear explanation of trust scores and warnings.

In [11]:

```python
# Visualize domain means using all available responses.
ax = domain_unpaired.set_index("domain")[["before_mean", "after_mean"]].plot(
    kind="bar",
    figsize=(9, 5),
    rot=35,
)
ax.set_ylim(1, 5)
ax.set_ylabel("Mean Likert score (1–5)")
ax.set_title("Domain–level descriptive comparison: all before vs. all after responses")
plt.tight_layout()
plt.show()
```

```python
# save image in output directory
OUTPUT_DIR = DATA_DIR / "survey_analysis_outputs"
OUTPUT_DIR.mkdir(exist_ok=True)
ax.get_figure().savefig(OUTPUT_DIR / "domain_means_all_responses.png", bbox_inches="tight")
```

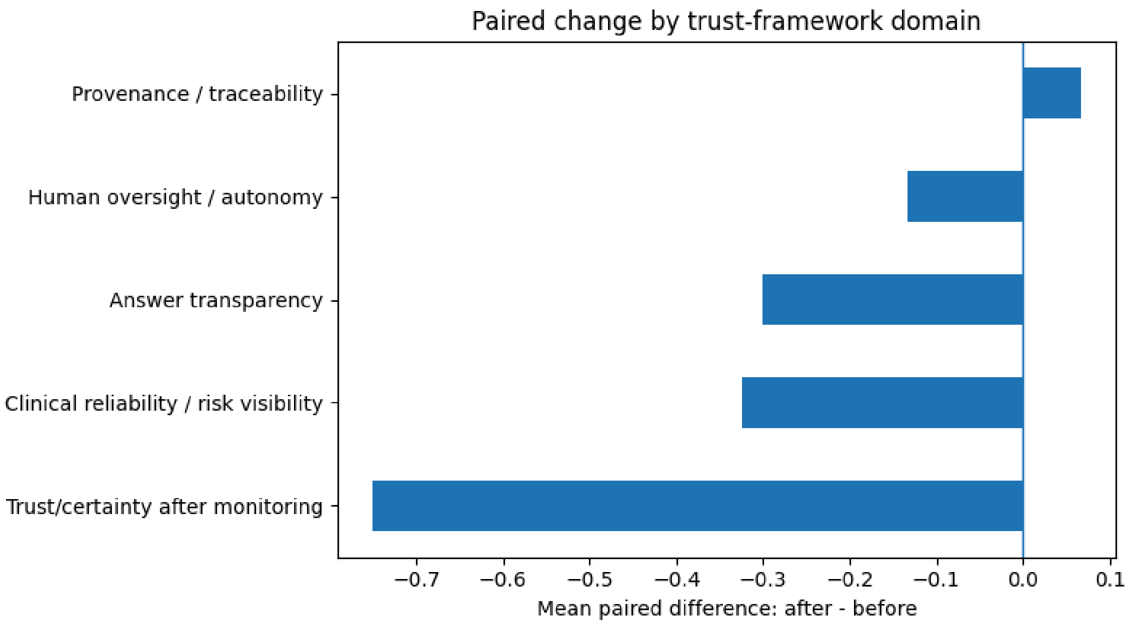


This figure compares average domain scores using all available baseline and follow-up responses. The visual confirms that most follow-up domain means are slightly lower than the baseline means, except for provenance and traceability, which remains roughly stable or slightly positive.

The result should be framed carefully. It does not support a claim that the prototype increased clinician trust. Instead, it suggests that the source and traceability mechanisms were the most successful part of the current implementation, while the broader answer presentation and trust indicators communication require refinement.

In [12]:
```
# Visualize paired domain differences.
plot_df = domain_paired.sort_values("paired_mean_diff")
ax = plot_df.plot.barh(
    x="domain",
    y="paired_mean_diff",
    figsize=(8, 4.5),
    legend=False,
)
ax.axvline(0, linewidth=1)
ax.set_xlabel("Mean paired difference: after - before")
ax.set_ylabel("")
ax.set_title("Paired change by trust-framework domain")
plt.tight_layout()
plt.show()

# save image in output directory
OUTPUT_DIR = DATA_DIR / "survey_analysis_outputs"
OUTPUT_DIR.mkdir(exist_ok=True)
ax.get_figure().savefig(OUTPUT_DIR / "paired_domain_differences.png", bbox_inches="tight")
```



This paired domain figure is one of the most important visuals in the notebook. It shows that the largest negative paired change is trust/certainty after trust indicators, while provenance and traceability is the only domain with a small positive change.

The practical interpretation is that participants may have noticed the provenance mechanisms, but the full trust framework was not yet communicated clearly enough to increase perceived trust or certainty. This aligns with the session notes, where participants valued citations, reliability indicators, warnings, and feedback, but found the meaning and presentation of these elements unclear or difficult to access.

In [13]:
```
AFTER_ONLY_FRAMEWORK_ITEMS = [
    "het systeem helpt mij om de gegenereerde antwoorden te beoordelen.",
    "de monitoring informatie heeft mijn vertrouwen in het systeem verhoogd.",
    "het systeem heeft me geholpen om onjuiste of onbetrouwbare antwoorden te ontdekken.",
```

```
    "het systeem heeft op een gepaste manier mijn vertrouwen verlaagd wanneer er onbetrouwbare of onjuiste antwoord
    "ik zou overwegen om het systeem te gebruiken in en klinische context/tijdens mijn werk.",
]

# Keep only items present in this export.
AFTER_ONLY_FRAMEWORK_ITEMS = [q for q in AFTER_ONLY_FRAMEWORK_ITEMS if q in after.columns]

after_framework = pd.DataFrame([
    {
        "question": q,
        "short_question": shorten(q),
        "n": int(after[q].map(to_likert_score).notna().sum()),
        "mean": after[q].map(to_likert_score).mean(),
        "median": after[q].map(to_likert_score).median(),
        "agree_or_strongly_agree_n": int((after[q].map(to_likert_score) >= 4).sum()),
        "neutral_n": int((after[q].map(to_likert_score) == 3).sum()),
        "disagree_or_strongly_disagree_n": int((after[q].map(to_likert_score) <= 2).sum()),
    }
    for q in AFTER_ONLY_FRAMEWORK_ITEMS
])

after_framework
```

Out[13]:

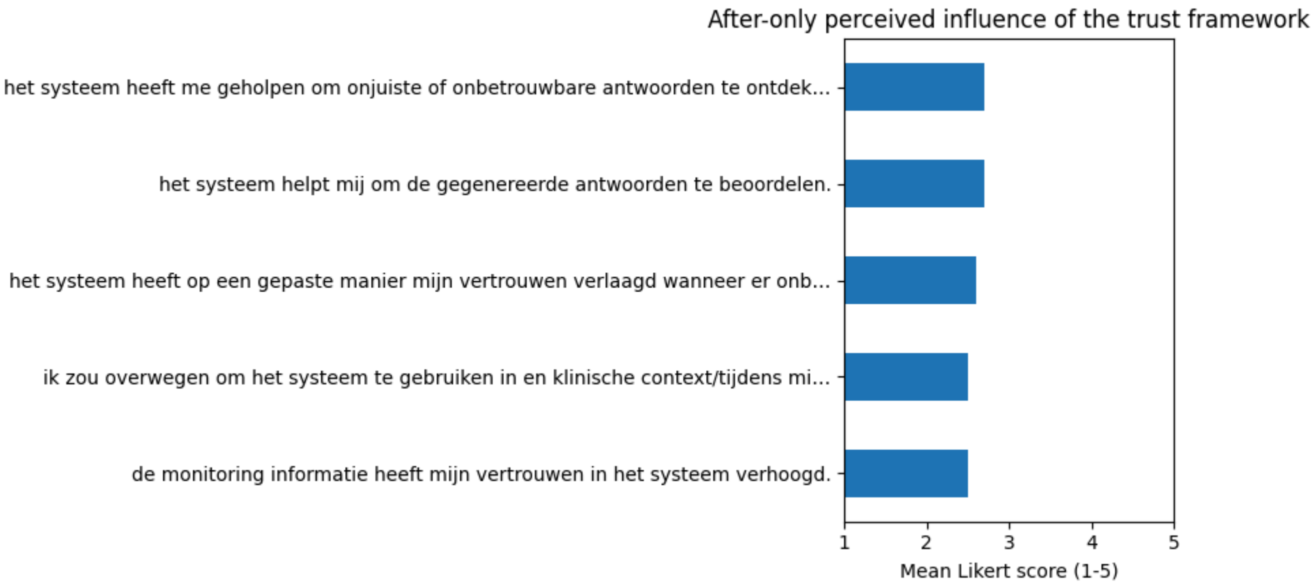
	question	short_question	n	mean	median	agree_or_strongly_agree_n	neutral_n	disagree_or_strongly_disagree_n
0	het systeem helpt mij om de gegenereerde antwoorden te beoordelen.	het systeem helpt mij om de gegenereerde antwoorden te beoordelen.	10	2.700000	3.000000	2	5	3
1	de monitoring informatie heeft mijn vertrouwen in het systeem verhoogd.	de monitoring informatie heeft mijn vertrouwen in het systeem verhoogd.	10	2.500000	3.000000	1	5	4
2	het systeem heeft me geholpen om onjuiste of onbetrouwbare antwoorden te ontdekken.	het systeem heeft me geholpen om onjuiste of onbetrouwbare antwoorden te ontdek...	10	2.700000	3.000000	1	6	3
3	het systeem heeft op een gepaste manier mijn vertrouwen verlaagd wanneer er onbetrouwbare of onjuiste antwoorden wer...	het systeem heeft op een gepaste manier mijn vertrouwen verlaagd wanneer er onb...	10	2.600000	3.000000	1	6	3
4	ik zou overwegen om het systeem te gebruiken in en klinische context/tijdens mijn werk.	ik zou overwegen om het systeem te gebruiken in en klinische context/tijdens mi...	10	2.500000	3.000000	2	4	4

These after-only questions ask participants directly about the perceived influence of the implemented trust framework after using the prototype. The means are between 2.5 and 2.7, with medians around neutral. This means that, quantitatively, participants did not strongly agree that the system helped them evaluate answers, increased trust, helped them detect unreliable answers, or would already be usable in clinical work.

This should be interpreted as a limitation of the current prototype rather than a rejection of the framework concept. The open answers and session notes show that participants saw value in the mechanisms, especially sources, reliability information, warnings, and feedback, but the implementation needs clearer communication and better presentation.

In [14]:
```
if len(after_framework):
    ax = after_framework.sort_values("mean").plot.barh(
        x="short_question",
        y="mean",
        figsize=(9, 4.5),
        legend=False,
    )
    ax.set_xlim(1, 5)
    ax.set_xlabel("Mean Likert score (1–5)")
    ax.set_ylabel("")
    ax.set_title("After-only perceived influence of the trust framework")
    plt.tight_layout()
    plt.show()

    # save image in output directory
    OUTPUT_DIR = DATA_DIR / "survey_analysis_outputs"
    OUTPUT_DIR.mkdir(exist_ok=True)
    ax.get_figure().savefig(OUTPUT_DIR / "after_only_framework_items.png", bbox_inches="tight")
```



The figure reinforces that the after-only framework ratings are below or around neutral. This means the trust mechanisms were not yet experienced as sufficiently clear, reliable, or clinically ready in the current prototype.

The main design implication is that the framework mechanisms should be made more understandable and actionable. For example, reliability scores need clearer terminology, citations should show more clearly which text supported the generated answer, warnings should explain what users should do next, and feedback should be integrated into the workflow without requiring researcher prompting.

In [15]:
```
MIN_PAIRED_N_FOR_TEST = 5

if len(paired_ids) < MIN_PAIRED_N_FOR_TEST:
    print(
        f"Statistical tests skipped: only {len(paired_ids)} paired respondents. "
        f"Use descriptive/formative interpretation until at least {MIN_PAIRED_N_FOR_TEST} paired responses are avai
    )
```

```python
else:
    from scipy.stats import wilcoxon

    test_rows = []
    b_idx = before_domains.set_index("participant_key")
    a_idx = after_domains.set_index("participant_key")

    for col in domain_cols:
        b = b_idx.loc[paired_ids, col]
        a = a_idx.loc[paired_ids, col]
        diff = (a - b).dropna()
        if len(diff) >= MIN_PAIRED_N_FOR_TEST and not np.allclose(diff, 0):
            stat, p = wilcoxon(diff)
        else:
            stat, p = np.nan, np.nan
        test_rows.append({
            "domain": col.replace("domain::", ""),
            "paired_n": len(diff),
            "wilcoxon_statistic": stat,
            "p_value": p,
        })

    statistical_tests = pd.DataFrame(test_rows)
    display(statistical_tests)
```

	domain	paired_n	wilcoxon_statistic	p_value
0	Answer transparency	10	11.500000	0.234375
1	Provenance / traceability	10	6.500000	0.875000
2	Clinical reliability / risk visibility	10	9.000000	0.250000
3	Human oversight / autonomy	10	13.500000	0.609375
4	Trust/certainty after monitoring	10	0.000000	0.031250

The Wilcoxon signed-rank test was used as an exploratory paired test for the domain scores. Most domains do not show statistically significant before/after differences. The exception is trust/certainty after trust indicators, where the p-value is below 0.05, but the direction of the change is negative rather than positive.

This result should be interpreted cautiously. The sample size is small, the analysis includes multiple exploratory domain tests, and the validation was formative rather than confirmatory. Therefore, the result should not be used to claim a proven statistical effect. Instead, it supports the descriptive observation that participants became less certain or less trusting after seeing trust related information. This may reflect trust calibration, but it may also reflect prototype limitations such as poor answer quality, unclear scores, or confusing UI presentation.

In [16]:
```
qualitative_cols = [
    "wat was uw eerste indruk van de tool?",
    "wat viel u positief op aan het uiterlijk van de tool?",
    "wat vond u onhandig of onduidelijk aan de tool?",
    "welke monitoring informatie heeft het meest geholpen bij het evalueren van een antwoord?",
    "en waarom?",
    "heeft u nog andere opmerkingen of vragen?",
]
qualitative_cols = [c for c in qualitative_cols if c in after.columns]

qualitative_after = after[qualitative_cols].copy()
qualitative_after.index = [f"After respondent {i+1}" for i in range(len(qualitative_after))]
qualitative_after
```

Out[16]:

	wat was uw eerste indruk van de tool?	wat viel u positief op aan het uiterlijk van de tool?	wat vond u onhandig of onduidelijk aan de tool?	welke monitoring informatie heeft het meest geholpen bij het evalueren van een antwoord?	en waarom?	hee opme of v
After respondent 1	Sceptisch	Duidelijke en rustige lay-out	Dat een juiste vraagstelling je sneller bij het antwoord brengt	Uitleg bij het antwoord;Betrouwbaarheid;	Leuk om een nieuwe toonl te krijgen want met zenya moet je heel vaak zoeken en de juiste woorden erin zetten en soms...	
After respondent 2	Leuk idea maar nog niet goed getraind	Simpel	Onduidelijk in een oog opslag hoeveel die nou van een bron gebruikt	Betrouwbaarheid;Uitleg bij het antwoord;	Er zijn nog verbeterpunten maar het idea is goed	
After respondent 3	Overzichtelijk	Simpel	Niet het antwoord wat ik hoop	Bronnen;	Omdat ik graag meteen het protocol wil zien	
After respondent 4	Slecht. Het generatieve aspect vind ik onnodig daar het zoektijd onnodig lang maakt, veel foute inhoud heeft en ik w...	Vriendelijk voor de ogen	Weinig context bij vragen	Bronnen;Waarschuwingen;	Reeds beantwoord	
After respondent 5	Lijkt me handig	Rustig en overzichtelijk	Vreemde antwoorden	Betrouwbaarheid;Uitleg bij het antwoord;	Kwaliteit van het antwoord	
After respondent 6	Op zich heel positief. Ik vind het idee hierachter erg goed en ik ben	Het ziet er makkelijk en professioneel uit.	Op dit moment niets.	Bronnen;Betrouwbaarheid;	De tool is prettig, maar er moet veel verbetering komen mbt de kwaliteit van de	Ik ber te over vanv kwal

	wat was uw eerste indruk van de tool?	wat viel u positief op aan het uiterlijk van de tool?	wat vond u onhandig of onduidelijk aan de tool?	welke monitoring informatie heeft het meest geholpen bij het evalueren van een antwoord?	en waarom?	hee opme of v
	er voorstander van.				antwoorden/bronnen en de snelheid va...	antw Echte voor
After respondent 7	Overzichtelijk	Duidelijk	Niets	Bronnen;	Er kwam verschillende info naar boven en liep door elkaar.	
After respondent 8	Het idee er achter is fijn	Het is overzichtelijk	De tekst klopt niet en voelde niet betrouwbaar	Bronnen;	Meerdere protocollen leken op elkaar in verweven te worden.	
After respondent 9	Tool is overzichtelijk	Rustig scherm	Om de juiste vraag te stellen om bij het protocol te komen	Uitleg bij het antwoord;	Hopelijk vanuit de testfase gaat hier \nverbetering in komen	Wee dit helpen lang antv
After respondent 10	Indrukwekkend en goed idee, als het werkt.	Makkelijk te gebruiken	De antwoorden en uitkomsten, de bronnen en betrouwbaarheid vond ik wel handig maar oogde wat onrustif omdat het midd...	Bronnen;Betrouwbaarheid;Waarschuwingen;Uitleg bij het antwoord;	Het is heel handig als het vlot en duidelijk werkt. Vooral wennen denk ik. \n\n	

The qualitative responses are essential for understanding the quantitative results. The open answers show that participants were critical of the current prototype, especially regarding answer quality, unclear or dense answer presentation, language issues, speed, and uncertainty about how much of the answer came from which source. These issues help explain why the Likert scores did not improve.

At the same time, the qualitative data is more optimistic about the underlying idea and the trust mechanisms. Participants mentioned sources, explanations, reliability information, and warnings as useful for evaluating answers. The accompanying group-session notes show a similar pattern, several participants saw potential in the system, participants looked first at citations and later at reliability information, and feedback became routine after prompting. Therefore, the validation should be interpreted as mixed but constructive, the trust mechanisms appear relevant, but their communication and UI integration require further refinement.

In [17]:
```
OUTPUT_DIR = DATA_DIR / "survey_analysis_outputs"
OUTPUT_DIR.mkdir(exist_ok=True)

question_summary.to_csv(OUTPUT_DIR / "question_level_unpaired_summary.csv", index=False)
paired_question_results.to_csv(OUTPUT_DIR / "question_level_paired_summary.csv", index=False)
domain_unpaired.to_csv(OUTPUT_DIR / "domain_unpaired_summary.csv", index=False)
domain_paired.to_csv(OUTPUT_DIR / "domain_paired_summary.csv", index=False)
after_framework.to_csv(OUTPUT_DIR / "after_only_framework_items.csv", index=False)
qualitative_after.to_csv(OUTPUT_DIR / "after_qualitative_notes.csv", index=True)
statistical_tests.to_csv(OUTPUT_DIR / "paired_domain_statistical_tests.csv", index=False)

print(f"Saved analysis tables to: {OUTPUT_DIR}")
```

```
Saved analysis tables to: data/survey_analysis_outputs
```

Discussion

Taken together, the results provide a mixed but useful formative validation picture. The quantitative Likert results do not demonstrate that the current prototype increases clinician trust. Most paired domain changes are neutral or negative, and the only clearly positive quantitative pattern is related to source visibility and provenance. The clearest negative pattern is reduced trust/certainty after seeing monitoring information.

This should not be interpreted as a simple failure of the trust framework. In clinical AI, the objective is not to maximize trust, but to support **calibrated trust**: clinicians should trust well-grounded answers more, but become cautious when answers are uncertain, poorly supported, or risky. The qualitative findings suggest that participants did see value in trust mechanisms such as citations, reliability information, warnings, and feedback. However, they also showed that these mechanisms were not yet communicated clearly enough and were affected by prototype limitations such as answer generation quality, response time, mixed language output, and dense answer presentation.

The main conclusion is therefore that the validation does not prove effectiveness, but it does support the relevance of the framework. The trust mechanisms are perceived as potentially valuable, while the prototype requires refinement in answer quality, answer structure, terminology, source presentation, warning communication, and feedback integration. Future validation should test the refined framework against a baseline interface without trust mechanisms to determine whether it improves trust calibration rather than simply increasing or decreasing trust.

C Formative Validation Results: Group Validation with Nurses

Validation Plan

Enhancing Clinician Trust through a Trust-Enabling Framework for Clinical RAG-Based LLM Assistants

Fontys Master of Applied IT & Catharina Ziekenhuis Eindhoven

Introduction

The pilot user test will be conducted to evaluate the prototype in a realistic nursing context and to explore how nurses interact with its trust-supporting mechanisms. The test aims to assess whether participants can effectively use the system to retrieve and interpret protocol information, inspect and evaluate supporting sources, provide feedback on system responses, and recognize situations in which the available information is insufficient and additional verification or clinical judgment is required.

The evaluation will be conducted as a formative pilot study in a controlled setting outside the direct patient care environment. Participants will attend the session in a separate room during time allocated for a clinical teaching session. The prototype will be introduced as a proof-of-concept/prototype: a supportive AI-based tool designed to assist nurses in finding and interpreting protocol information, while leaving clinical judgement and decision-making to the user.

Session Structure

Activity	Time	Description
Pre-test questionnaire	5 min (filled in before test session)	Short questionnaire about background, AI experience, and attitude toward AI in a clinical context. Completion is checked before the test starts.
Pilot user test	20 min	Participants use the prototype individually and simultaneously in a separate room outside the real care environment. They first receive a quick explanation about the system.
Post-test questionnaire	5 – 10 min	Questionnaire about experience, trust, usability, source visibility, and usefulness of the prototype. Follow-up questions are asked if time remains.

General Procedure Flow

- Participants are nurses taking part during time reserved for a clinical teaching session.
- Each session is expected to include approximately four to seven participants.
- Participants work individually while the group progresses through the scenarios one by one.
- Participants raise their hand when they finish a scenario, so the coordinator can manage the session flow.
- Observations are used to analyse interaction patterns afterwards.

- After scenario 1, participants are asked to test the feedback element. After scenario 2, they are asked whether enough information was found, to see whether they inspect the sources.

Scenarios

Scenario 1 – Specific Catheter Question

Case: Henk de Boer, 67, has 700 ml residual urine after micturition. The participant wants to insert a catheter and must find the correct protocol.

Expected protocol: 7699550, because it explicitly concerns a verblijfskatheter and the patient remains admitted on the ward.

Relevant but not final protocols: 7699548, eenmalige katheter; 7699545, eenmalige katheter vrouwen.

Tests: retrieval precision, correct protocol selection, source inspection, feedback use, and response to clarification needs such as catheter type, sex, adult/child, and placement context.

Scenario 2 – Complex Department Specific Question

Case: a 78-year-old orthopaedic patient with a petrochanteric femur fracture receives zoledronic acid after surgery. The participant must find the conditions before administration, how the medicine should be administered, and which side effects to monitor.

Expected protocol: 7705240 - Zoledroninezuur, version 4.

Relevant but not final protocols: 7705260, 7706075, and 7719135.

Tests: handling of a complicated clinical case, source inspection, interpretation of protocol conditions, and whether the participant can judge whether enough information has been found.

Scenario 3 – Question that cannot be answered directly

Case: the participant asks how to insert a vlindernaaldje.

Expected system behaviour: the system should not over-answer based only on the term. It should guide the user toward clarification, for example whether another name is used, such as Saf-T-Intima, and what the needle is used for, such as subcutaneous administration.

Relevant protocols: 7718538 may contain the answer but does not mention the term vlindernaaldje; 7717526 may link to the correct protocol; 7700261 may be expected but is not correct because the action is not described there.

Tests: clarification behaviour, safe handling of missing or indirect information, user reformulation, and avoidance of unsupported answers.

Analysis Focus

The results should be interpreted descriptively and formatively. The aim is not to prove clinical effectiveness, but to evaluate how the prototype supports trust calibration, transparency, source-based verification, and safe human oversight during protocol retrieval.

The analysis focuses on:

- whether participants can find the correct protocol or follow a plausible and clinically meaningful path toward it
- whether participants inspect sources when source information is available, especially in situations where multiple related protocols may appear relevant
- whether source visibility helps participants understand where an answer comes from and whether the answer is sufficiently supported
- whether participants use the feedback element and whether this mechanism is clear enough to support human oversight and future system improvement
- whether participants recognize when clarification is needed, when terminology is ambiguous, or when the available information is incomplete
- whether the prototype helps participants distinguish between a reliable answer, a partially supported answer, and a situation where further verification is required
- whether participants maintain appropriate trust in the system, accepting well-supported answers while remaining critical of uncertain, incomplete, or potentially unsupported information
- whether the prototype encourages safe decision-making behaviour by supporting verification, escalation, and clinical judgement rather than replacing the nurse’s responsibility
- whether warning cues, explanations, source information, and reliability-related signals are understandable and useful in practice
- whether observed usability issues, unclear terminology, or interaction problems reduce transparency, trust calibration, or the ability to verify the answer safely

Enhancing Clinician Trust through a Trust-Enabling Framework for Clinical RAG-Based LLM Assistants

D Prototype Implementation of the Clinician Trust Framework

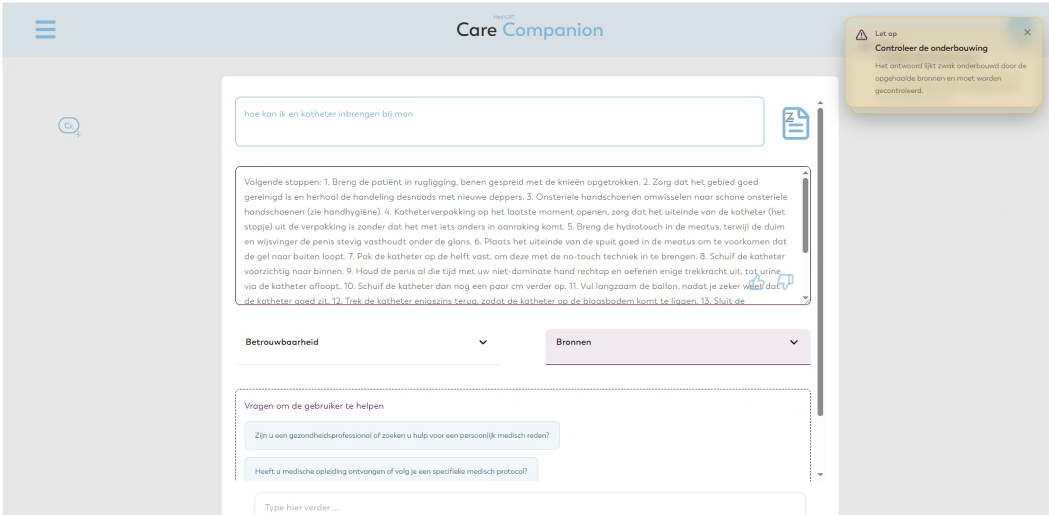


Figure 6: Example of a question-answering interaction with the clinical RAG-based LLM assistant. The interface shows the generated answer, retrieved sources, reliability information, warnings, follow-up questions/suggestions, and feedback options to support clinician trust and informed judgement.

E Ethical Evaluation of the Clinician Trust Framework

Ethical Assessment

Enhancing Clinician Trust through Structured Monitoring and Governance of Clinical RAG-Based Large Language Models

Fontys Master of Applied IT & Catharina Ziekenhuis Eindhoven

This document examines the ethical implications of an evaluation & monitoring and governance framework for clinical RAG-based LLM assistants. Within the context of the research project, the purpose of this document is to identify ethical risks, societal concerns, and governance requirements at an early stage, so that these insights can directly inform the design, validation, and responsible development of the proof-of-concept and the broader framework. It therefore serves not only as an ethical reflection, but also as a design-oriented input for the research project.

Introduction

In clinical settings, trust in retrieval-augmented generation (RAG)–based large language model (LLM) assistants is constrained by the absence of robust monitoring and governance frameworks [1], [2]. Although these systems can support healthcare professionals by enabling faster access to relevant information, their deployment introduces unresolved concerns related to output correctness, hallucination risk, safety, uncertainty handling, drift, provenance, and accountability [3], [4], [5].

The increasing integration of RAG-based LLM assistants into clinical workflows amplifies the importance of addressing these challenges. As a result, clinicians require systems that not only provide useful information but also offer clear insight into how outputs are generated, which sources are retrieved, and how reliability is ensured over time [6], [7].

Despite growing research on large language models in healthcare, existing approaches primarily focus on technical performance and model capabilities [4]. Comparatively less attention has been given to the design of structured monitoring and governance frameworks that integrate transparency, traceability, and accountability into everyday clinical use [2], [5], [8]. This creates a gap between conceptual discussions of AI risks and the practical implementation of mechanisms that support safe and trustworthy deployment in real-world hospital settings.

This problem affects multiple stakeholders, including clinicians, patients, hospital governance bodies, and system developers. Without adequate monitoring and governance, clinicians are unlikely and unwanted to rely on AI systems that cannot explain their outputs or demonstrate reliability [1], [6], [7]. Consequently, the lack of oversight mechanisms may hinder responsible adoption, limit compliance with regulatory requirements, and reduce trust in AI-assisted clinical decision-making.

Methodology

The ethical assessment uses a qualitative, exploratory approach to examine the ethical, social, and practical implications of using retrieval-augmented generation (RAG)–based large language models in healthcare contexts. This assessment forms part of the socio-technical evaluation stage within the Design Science Research (DSR) methodology, contributing to the validation of the proposed monitoring and governance framework

The Technology Impact Cycle Tool (TICT) was employed as the primary analytical framework [9]. TICT enables reflection on the societal and ethical implications of technology through domain-specific guiding questions. The responses were synthesised into themes related to impacts, risks, and opportunities, establishing a theoretically grounded basis for subsequent design and evaluation activities.

To support iterative refinement, an expert discussion session was conducted with a professional ethicist and a legal expert. This activity aligns with the contextualisation and evaluation phases of the DSR process, ensuring that the emerging themes reflect both ethical theory and regulatory requirements. Attention was given to ethical principles such as responsibility, autonomy, and transparency, as well as regulatory considerations including data protection, liability, and compliance obligations. Insights from this session informed the iterative development of the ethical framework and its integration into the broader governance design.

Finally, a multidisciplinary brainstorming session was organised with professionals involved in the design and implementation of RAG-based LLM assistant within a hospital context. Participants included an industrial design master's student specialising in UI/UX design, a data scientist responsible for data preparation and provisioning, and a project manager with direct insight into clinical workflows and clinician needs.

Using the Rose, Thorn, Bud method [10], the group analysed three scenarios involving a nurse interacting with a RAG-based LLM assistant. This helped identify strengths (roses), challenges (thorns), and future opportunities (buds), supporting the translation of ethical considerations into actionable design and governance requirements.

Results

Legal Implications

A clinical RAG-based LLM assistant may be misused, directly or indirectly violate legal, regulatory, or institutional requirements. In addition, weak provenance and logging mechanisms could allow users or organizations to avoid accountability by making it difficult to reconstruct which prompts, retrieved sources, model versions, or human edits contributed to a specific output [8], [11]. This is particularly important in healthcare, where transparency, auditability, risk management, and post-deployment monitoring are essential requirements for trustworthy and compliant AI use [2], [5], [12]. Therefore, the legal and governance risk arises from insufficient monitoring and documentation, which may enable non-compliant use while obscuring responsibility after incidents.

Privacy

In the current context, the RAG-based LLM assistant will have access only to medical protocols that do not contain personal data. However, personal data may still enter the system, either directly or indirectly, through clinician prompts, uploaded or retrieved documents, audit logs, and user feedback. In addition to patient-related data, the monitoring and governance framework may also process clinician-related data, including user identifiers, timestamps, review actions, edits, and override decisions [2], [11]. This issue is particularly significant as the research project emphasizes transparency, provenance, traceability, and logging, all of which may create additional privacy obligations when they involve identifiable indicators. Privacy should therefore be treated not as secondary, but as a core design constraint of the monitoring architecture [12].

Influence on Identity

A RAG-based LLM assistant can be empowering when it functions as a transparent and reviewable support tool that helps clinicians access relevant information efficiently. In such cases, a RAG-based LLM assistant enhances clinical decision-making while preserving the clinician's control and responsibility. However, a RAG-based LLM assistant may also threaten autonomy and dignity if it is perceived as a black-box authority, if users feel pressured to conform to its outputs, or if monitoring mechanisms are experienced as surveillance rather than support [1], [7]. In addition, the technology may alter relationships among clinicians and between clinicians and patients by changing how expertise, responsibility, and trust are perceived in clinical work. For this reason, the governance framework should not only monitor technical performance but also preserve human oversight and the clinician's role as the accountable decision-maker.

Data

It is recognized that data may be partial, contextual, and potentially biased rather than objective reflections of reality [4], [13]. This is highly relevant for a RAG-based LLM assistant, which depends on retrieved sources as well as on prompts, and user feedback. It is likely that sources are incomplete, outdated, or interpreted out of context [3], [4], [5]. It is important to focus on monitoring dimensions such as output correctness, hallucination risk, uncertainty behaviour, provenance, drift, and clinical feedback, but also to make transparent which data sources are searched and which information is not available to the system [3], [5], [11], [14]. This helps users understand what may be missing from an answer, for example when a newly introduced clinical protocol has not yet been added to the database, or when one source is disproportionately influential compared to others. It is essential to acknowledge the limitations of RAG-based LLM assistants and their underlying data sources, and to ensure structured oversight and insight.

Fairness & Bias

A RAG-based LLM assistant may contain built-in bias at multiple levels, including the underlying language model, the retrieval corpus, the evaluation process, and the monitoring framework itself. In a clinical setting, bias may emerge if certain patient groups, medical conditions, languages, documentation styles, or clinical workflows are better represented than others. This research project recognizes fairness as an important monitoring challenge. For that reason, this research project should not assume bias can be eliminated completely. Instead, bias should be treated as a persistent socio-technical risk that must be monitored through transparency, auditability, feedback mechanisms, and periodic review of both data sources and system behaviour [5], [13], [14].

Informing & Communication

Users and stakeholders should be able to understand which sources were used, which model and prompt version was used, and how errors, uncertainty, or drift are handled [3], [7]. It is necessary to explain AI rationale, provenance, model metadata, and links to technical documentation [6], [7], [11]. These are important mechanisms for transparency and trustworthiness. Therefore, a monitoring and governance framework should make the system operationally understandable. If external vendors or commercial models are involved, transparency about roles, responsibilities and dependencies is also necessary.

Sustainability

Energy use is a relevant but currently an out of scope consideration in this project. A clinical RAG-based LLM assistant may consume energy both directly, through model inference, retrieval, embedding updates, and monitoring processes, and indirectly, through storage, logging, infrastructure dependencies, and vendor-hosted services. Performance monitoring can help identify the components with the highest energy consumption and inform targeted optimization measures. Although sustainability is not the central focus of the project, it should be acknowledged as an important design consideration in future operational implementation.

Future

In the future, the monitoring framework ideally would become a routine part of clinical information work, especially if RAG-based LLM assistants are scaled across departments and integrated more deeply into hospital workflows. At larger scale, the most important issue may not be whether AI is used, but how its use becomes normalized, governed, and justified within healthcare institutions. This reinforces the relevance of this research project, which focuses on the conditions under which such systems can be monitored and governed responsibly rather than merely deployed.

Discussion

The ethical assessment highlights that the challenges associated with clinical RAG-based LLM assistants are not solely technical but fundamentally socio-technical in nature. The findings demonstrate that risks related to transparency, accountability, and trust emerge from the interaction between system design, data characteristics, clinical workflows, and governance structures [1], [2], [8]. As such, effective mitigation requires an integrated approach that combines technical monitoring with organisational and ethical oversight.

A central insight of this assessment is that transparency and traceability are not optional features but core conditions for responsible AI deployment in healthcare [11], [12]. The identified concerns regarding provenance, logging, and explainability indicate that clinicians require verifiable insight into how outputs are generated [1], [6], [7]. Without such mechanisms, the system risks being perceived as a black box, which undermines trust and limits adoption. This reinforces the importance of embedding auditability, model metadata, and retrieval visibility directly into the monitoring framework, rather than treating them as secondary features.

In addition, monitoring should extend beyond performance metrics to include ethical and behavioural dimensions [4], [5], [12]. Issues such as bias, data incompleteness, and uncertainty require continuous oversight, feedback loops, and explicit communication to users [5], [13], [14]. This aligns with the recognition that bias is a persistent condition rather than a problem that can be eliminated and must be actively monitored and contextualised within clinical use [5], [13].

While detailed logging and traceability are necessary for accountability and regulatory compliance, they may introduce additional privacy risks, particularly when clinician actions and potentially sensitive inputs are recorded [11], [12]. This highlights the need for privacy-by-design principles within the monitoring architecture, ensuring that data collection is proportionate, secure, and aligned with regulatory requirements. Balancing these competing demands is a key design challenge for governance frameworks in clinical AI systems.

Furthermore, the assessment underscores the importance of preserving clinician autonomy and professional responsibility. Although RAG-based LLM assistants can enhance efficiency and decision support, there is a risk that users may over-rely on system outputs or perceive the system as authoritative [1], [7]. This could shift responsibility in unintended ways and affect clinical judgement. The proposed framework should include human-in-the-loop principles, ensuring that clinicians remain the final decision-makers and that the system supports, rather than replaces, professional expertise.

From a Design Science Research perspective, these insights provide direct input for the design and refinement of the monitoring and governance framework. The identified ethical risks can be translated into concrete design requirements, such as the implementation of provenance tracking, uncertainty indicators, audit logs, feedback mechanisms, and bias monitoring processes [2], [5]. In this way, the ethical assessment contributes not only to reflection but also to actionable design decisions within the artifact development process.

Finally, the findings suggest that the long-term challenge is not only the safe deployment of RAG-based LLM assistants, but their normalization within clinical practice. As these systems become more integrated into workflows, governance mechanisms must scale accordingly and remain adaptable to evolving regulatory, technical, and organisational conditions. Continuous monitoring, stakeholder involvement, and iterative evaluation will therefore be essential to sustain trust over time.

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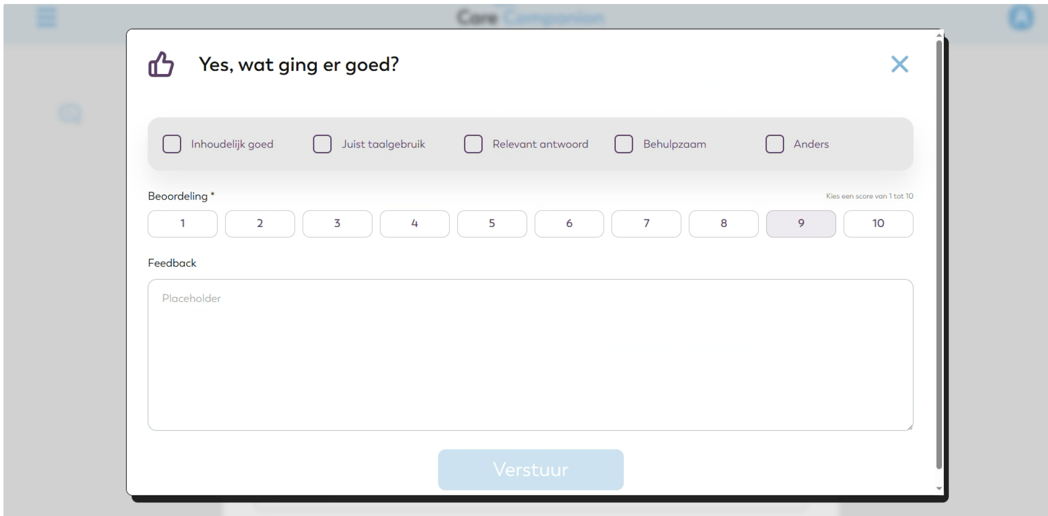

Figure 7: Example of a feedback capture interface in the clinical RAG-based LLM assistant. The interface allows clinicians to provide feedback on the generated answer, including rating its helpfulness, indicating agreement or disagreement, and providing comments.

Figure 8: Example of a monitoring dashboard in the clinical RAG-based LLM assistant. The dashboard provides an overview of the system's performance and the status of ongoing interactions.

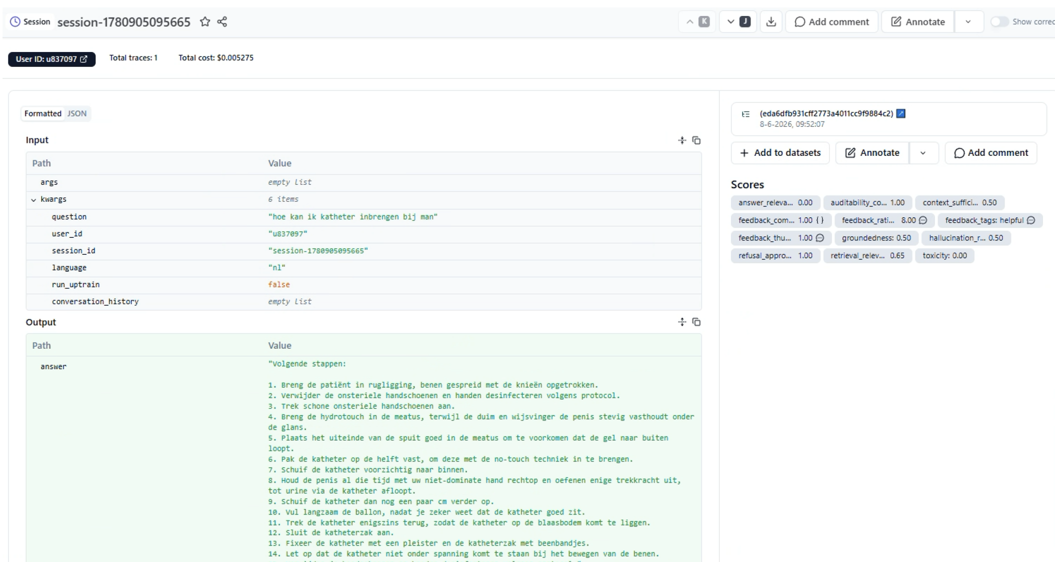

Figure 9: Example of a trace log in the clinical RAG-based LLM assistant. The trace log provides a detailed record of the system's interactions and decision-making process.